

Miniaturizing Optical Systems for Laser Characterization and Stabilisation in Quantum Applications

By

Aditya Tyagi
Aniket Londhe
Brandon Geater
Ed Wood
Hania Shahid
Oliver French
Priya Saini

MSc Optoelectronics and Quantum Technologies



Department of Electrical, Electronic, and Mechanical Engineering
UNIVERSITY OF BRISTOL

A MSc dissertation submitted to the University of Bristol in accordance with the requirements of the degree of MASTER OF SCIENCE IN OPTOELECTRONICS AND QUANTUM TECHNOLOGIES in the Faculty of Engineering.

September 11, 2026

Abstract

Optical systems used in quantum technologies often rely on expensive, stationary laboratory equipment, limiting portability and accessibility. This project investigates two approaches to miniaturisation: a low-cost portable free-space optical platform and a photonic integrated circuit (PIC).

The free-space system was constructed on a compact aluminium breadboard using predominantly 3D-printed optical mounts. Knife-edge laser characterisation produced a Gaussian fit of $R^2 = 0.995$ and a measured $1/e^2$ beam radius of 0.281 mm. Mechanical stability testing showed only approximately 0.03% variation in detected optical signal under repeated external perturbations. Michelson and Mach-Zehnder interferometry further demonstrated the platform's ability to reproduce standard optical experiments, including polarisation-dependent interference.

For further miniaturisation, a silicon nitride PIC operating at 679 nm was designed and simulated using dual asymmetric Mach-Zehnder interferometers with differing free spectral ranges. Circuit-level simulations assessed wavelength discrimination, noise-limited performance, and collision behaviour, yielding a collision-free operating range of approximately $679 \text{ nm} \pm 0.606 \text{ nm}$. Together, these approaches demonstrate a progression from portable free-space optics towards compact integrated photonics.

Declaration of own work

This project report is submitted towards an application for a degree at the University of Bristol. The report is based upon independent work by the candidates. All contributions from others have been acknowledged and the supervisor is identified on the front page. The views expressed within the report are those of the authors and not of the University of Bristol. We hereby assert our right to be identified as the authors of this report. We give permission to the University of Bristol Library to add this report to its stock and to make it available for consultation in the library, and for inter-library lending for use in another library. It may be copied in full or in part for any bona fide library or research worker on the understanding that users are made aware of their obligations under copyright legislation. We hereby declare that the above statements are true.

Aditya Tyagi
Aniket Londhe
Brandon Geater
Ed Wood
Hania Shahid
Oliver French
Priya Saini

© Copyright, Aditya Tyagi, Aniket Londhe, Brandon Geater, Ed Wood, Hania Shahid, Oliver French, Priya Saini, 2026 Certification of ownership of the copyright in a dissertation presented as part of and in accordance with the requirements for the relevant degree at the University of Bristol. This report is the property of the University of Bristol Library and may only be used with due regard to the authors. Bibliographical references may be noted but no part may be copied for use or quotation in any published work without prior permission of the authors. In addition, due acknowledgement for any use must be made.

Work Allocation

Name	Tasks Done
Aditya Tyagi	Procured components, assembled, and aligned optical setups across multiple experimental configurations (Michelson, MZI, Quantum Erasure), including a portable version. Developed setup circuitry for Arduino logging, thermistor/photodiode measurement logging code, and Peltier/TEC power and error analysis. Performed fringe spacing analysis, prepared LaTeX figures and plots, formatted the report, and coordinated team deliverables.
Aniket Londhe	Designed and optimized initial and final Photonic Integrated Circuit (PIC) interconnect models based on geometry and S-parameters. Conducted FDTD and EME simulations using Ansys Lumerical on HPC to maximize light transmission and evaluate Figures of Merit (FOM). Produced final PIC layout designs in KLayout and GDSFactory using actual component measurements.
Brandon Geater	Designed, 3D-printed, and assembled extensive optical and mechanical hardware, including kinematic mirror mounts, mirror capsules, posts, polariser mounts/capsules, lens mounts/capsules, laser clamps, breadboard mounts, irises, and enclosure box components. Created technical engineering drawings for the University of Bristol machining workshop to fabricate the custom breadboard and laser mount.
Ed Wood	Designed, simulated, and extracted S-parameters for all prototype and final PIC components, building full Interconnect models for subsystem and circuit-level performance validation. Wrote Lumerical scripts, Excel formulas, and VBA macros to automate data extraction and error signal analysis. Documented control system architectures and specifications, managed the Gantt chart, and coordinated team meetings.
Hania Shahid	Designed and assembled the free-space experimental framework, Michelson and Mach-Zehnder interferometers, Quantum Erasure setup, and mechanical vibration damping. Developed system circuitry for Arduino logging, performed knife-edge measurements to reconstruct transverse Gaussian intensity profiles, created 3D-printed mounts for beam splitters, developed evaluation methodology, and co-managed report formatting and team deliverables.
Oliver French	Researched suitable lasers, studied locking techniques, and designed temperature control schemes, including circuit design. Informed laser mount/clamp designs, assembled and tested the TEC, completed Steinhart-Hart thermistor modeling, and characterized laser temperature responses. Assisted with HPC access, interconnect S-parameter modeling, and knife-edge data collection.
Priya Saini	Conducted literature studies on PIC structures and SiN edge couplers, and designed 3D mirror/lens mount holders. Re-configured PIC components to extract S-parameters across waveguides, 2x2MMI, and 1x2MMI structures using FDE, EME, and FDTD solvers. Designed edge couplers with inverse tapers for mode matching and evaluated Figures of Merit for power, loss, and phase imbalance.

Contents

1	Introduction	4
2	Theoretical Background - AT	6
2.1	Light and Coherence	6
2.2	Polarization	6
2.3	Laser Characterization	7
2.4	Interferometry	7
2.5	Optical Atomic Clocks	8
3	Free-Space Setup	9
3.1	Research Methodology - HS, BG	9
3.2	Temperature control attempt - OF	16
3.3	Results - HS	18
3.4	Discussion - HS	27
4	Photonic Integrated Circuit	31
4.1	Research Methodology - EW	31
4.2	Design - EW	33
4.3	Components - PS	36
4.4	Results - EW	43
4.5	Discussion - EW	51
4.6	Future Work - EW	52
5	Conclusion	54
A	Appendix	56
A.1	Electronic components	56
A.2	Voltage Vs Polarization Plot	56
A.3	Summary of resistor changes for MAX1978 board	57
	References	58

1 Introduction

Photonic systems are integral to modern communication systems, sensing technologies, quantum technologies and pedagogical research in optics. In the context of this report, optical atomic clocks and related quantum applications have been considered. The standard hardware required for such devices is often bulky and expensive; therefore, to address these limitations, this research first investigates the performances development of a low-cost and portable free-space optical system intended for pedagogical and research applications, before progressing to a more industry-standard photonic integrated circuit design and simulation. Due to time constraints, the latter is presented as a proof of concept for a ready-to-fabricate chip.

Conventional free-space optical experiments typically rely on precision optical tables, translation stages, commercial mounts, and discrete optical components. While such systems provide excellent alignment control and mechanical stability, their physical footprint and cost can restrict their use outside dedicated laboratory environments. Miniaturisation therefore presents an opportunity to develop more accessible and portable systems while retaining the functionality required to perform meaningful optical measurements. A central challenge, however, is determining the extent to which reductions in size and cost can be achieved without significantly compromising optical alignment, mechanical stability, repeatability, or experimental performance.

The first part of this work addresses this problem through the design and fabrication of a compact free-space optical platform. A miniature optical breadboard and a series of custom optical mounts were developed to accommodate conventional components including mirrors, beam splitters, lenses, polarizers, and a semiconductor laser source. The system was designed with an emphasis on low cost, portability, modularity, and mechanical stability. Its performance was subsequently evaluated through a series of experiments intended both to characterize the optical source and to determine whether fundamental laboratory-scale optical experiments could be reproduced on the reduced platform. Laser beam profiling using the knife-edge technique and interferometric experiments based on Michelson and Mach-Zehnder configurations were used to assess the validity and performance of the scaled-down setup.

While the free-space system demonstrates miniaturization through compact discrete optics, photonic integration provides a route towards substantially smaller, more stable, and scalable optical systems by implementing similar methods but with added complexity. The second part of this work therefore investigates the design and simulation of a silicon nitride photonic integrated circuit intended for wavelength discrimination and stabilization around 679 nm, corresponding to a repumping transition used in strontium optical atomic clocks¹. The broader objective of this work is to establish a complete design pathway from the optimisation of individual photonic building blocks to their integration into a functional wavelength-stabilization architecture. This includes the design and characterization of waveguides, bends, tapers and multimode interference couplers, followed by their incorporation into two asymmetric Mach-Zehnder interferometers with differing free spectral ranges. The dual-AMZI architecture is intended to generate complementary normalised error signals that provide sensitivity to wavelength variations, identify the direction of wavelength drift, and uniquely determine the operating wavelength over a wider range than would be possible using a single interferometer. The completed circuit is subsequently evaluated in terms of its wavelength discrimination range, possible wavelength ambiguities, sensitivity, and robustness to detector noise, thereby assessing whether the proposed architecture can provide a practical basis for laser-frequency stabilization. Although fabrication was beyond the available project timeframe, the PIC study aims to demonstrate a complete simulated proof of concept, progressing from individual component design through system-level validation towards a layout suitable for future fabrication and experimental testing.

The performance of a laser system depends firstly on the semiconductor diode source and then on the optical, electrical and thermal environment around it. The beam emitted from the diode is usually asymmetric and divergent, requiring shaping and collimation². Intensity, frequency and phase noise, as well as longer-term drift, can arise from electrical, thermal and mechanical (vibrational) fluctuations, and are therefore reduced through a combination of low-noise power, temperature stabilisation, mechanical isolation and active laser-locking techniques in which deviations from a chosen frequency reference are converted into an error signal and corrected³. For many laser

applications, especially optical atomic clocks, the laser must be stable with a narrow linewidth and so these locking and purification techniques are pertinent³. In order to create a laser system suitable for industrial applications a combination of these techniques tend to be used.

Control systems were developed for both the free-space and photonic integrated circuit designs in order to optimise the systems while obeying their prioritised aims (miniaturisation for the PIC and cost-effectiveness for the free space system). The photonic integrated circuit facilitates high-precision frequency discrimination for use in frequency locking. Due to cost and constraints arising from working with a class II laser, a frequency locking system was not possible for the free space system so an attempt was made to implement a temperature stabilisation system with a passive heat sink, rather than an active one, allowing the system to remain uncrowded to facilitate effective science communication. Unfortunately, initial interrogation of this set-up showed that thermo-electric control is not possible with a passive heat sink however, characterisation of the laser module's temperature and replication of fundamental quantum experiments indicated that the temperature stabilisation is superfluous for this free-space set up with a class II laser.

The overall aim of this investigation is therefore to examine two complementary approaches to the miniaturization and cost-reduction of optical systems while assessing the extent to which useful optical functionality can be preserved. Together, the two approaches provide a progression from conventional laboratory optics towards increasingly compact implementations.

2 Theoretical Background - AT

Optical atomic clocks are crucial to precision metrology, having the potential to redefine the SI second⁴. They utilise ultra-stable and high precision lasers to cause a narrow electronic transition in an atom like Sr and Yb^{5,1}. Optical atomic clocks and this report's fundamental concepts consist of light and its components, the characterisation of a laser, interferometry, and finally how they combine to make an optical atomic clock. Unless otherwise stated, "light" refers to the output of the laser used as the primary optical source in these experiments.

2.1 Light and Coherence

Light is an electromagnetic field that can be modelled as an oscillating wave. For a monochromatic plane wave, the electric field can be represented as:

$$E(z, t) = E_0 \cos(kz - \omega t + \phi), \quad (2.1)$$

where E_0 is the amplitude of the electric field, k is the wavevector, ω is the angular frequency and ϕ is the phase⁶.

Phase is where in the cyclical point of the wave it is at, i.e at the crest or the trough. Amplitude is defined as the maximum strength of the oscillation. Wavelength is the distance between one crest of the wave and the next crest. Polarization is the definition of the orientation of the oscillations in the electric field, i.e a wave can either oscillate along the x-axis or the y-axis, which is referred to as linearly polarized. The wavevector is a vector indicating the direction of wave propagation and the phase delay per unit length. λ describes the spatial period of the wave.

When discussing light, coherence is a core focus, describing the degree to which amplitude and phase are correlated in either space, time, or both⁷. It is significant in interferometry, where a phase relationship between two optical fields is required to produce a stable interference pattern. It is split into two kinds of coherence, spatial and temporal. Spatial coherence is the degree of correlation between electric fields at different transverse positions across a beam. Temporal coherence is the degree of correlation between the electric field at one point in space but different times. Temporal coherence is defined as follows:

$$\ell_c = \frac{\bar{\lambda}^2}{\Delta\lambda}, \quad (2.2)$$

where $\bar{\lambda}$ is the central wavelength of the light, $\Delta\lambda$ is the linewidth of the light, and ℓ_c is the coherence length⁸. Therefore, a narrower laser linewidth corresponds to a longer coherence length and a more stable phase relationship over a given propagation distance.

2.2 Polarization

Polarization describes the orientation of the electric field of an electromagnetic wave relative to its direction of propagation. Linear polarization occurs when the electric field oscillates along a fixed direction, while circular polarization occurs when two perpendicular electric-field components have equal amplitudes and a phase difference of $\pi/2$ ⁹. For example, the intensity transmitted through an ideal linear polarizer (I) is described by Malus' law,

$$I = I_0 \cos^2 \theta, \quad (2.3)$$

where I_0 is the incident intensity and θ is the angle between the polarization direction of the incident light and the axis of the polarizer¹⁰. Polarization is important in interferometry because two beams with orthogonal polarization states do not produce the same observable interference as beams with parallel polarizations.

2.3 Laser Characterization

The performance of a laser can be characterised through several parameters, including its wavelength, optical power, spectral linewidth, beam profile and frequency stability¹¹.

The spectral linewidth $\Delta\nu$ describes the range of frequencies over which the laser emits and is an important parameter when considering coherence. A narrower linewidth results in a longer temporal coherence length and therefore allows a stable phase relationship to be maintained over a greater optical path difference. The optical intensity of a beam is related to its optical power by

$$I = \frac{P}{A}, \quad (2.4)$$

where P is the optical power and A is the illuminated area⁶. For an ideal Gaussian beam, the transverse intensity distribution can be approximated by

$$I(r) = I_0 \exp\left(-\frac{2r^2}{w^2}\right), \quad (2.5)$$

where I_0 is the peak intensity, r is the radial distance from the beam axis and w is the beam radius at which the intensity falls to $1/e^2$ of its peak value¹⁰. These parameters are important when characterising a laser for an optical system, as changes in wavelength, power, beam profile or frequency stability can influence the performance of subsequent optical components and interferometric measurements.

2.4 Interferometry

Interferometry is a technique that uses the superposition of two or more coherent optical fields to measure differences in phase, optical path length or other physical quantities that affect the optical phase¹². Consider two electric fields with a phase difference $\Delta\phi$. When the fields are superimposed, the resulting intensity is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\phi), \quad (2.6)$$

where I_1 and I_2 are the intensities of the two fields⁶. The measured intensity therefore varies according to the phase difference between the two beams. The phase difference can be related to the optical path difference ΔL by⁶

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta L \quad (2.7)$$

2.4.1 Michelson Interferometer

A Michelson interferometer consists of a beam splitter, two mirrors and a detector. The incident laser beam is divided by the beam splitter into two separate arms, with each beam travelling to a mirror before being reflected back towards the beam splitter. The two beams are then recombined, producing an interference pattern at the output. The intensity measured at the detector therefore depends on the phase difference between the two arms. If one of the mirrors is displaced by a distance d , the optical path length changes by twice this distance because the beam travels to the mirror and returns. Therefore,

$$\Delta L = 2d, \quad (2.8)$$

and the corresponding phase change is⁶

$$\Delta\phi = \frac{2\pi}{\lambda} (2d) = \frac{4\pi d}{\lambda} \quad (2.9)$$

Consequently, very small changes in mirror position can produce measurable changes in the interference intensity. This sensitivity makes the Michelson interferometer useful for measuring small displacements and phase changes, and it provides a method of characterising the coherence properties and stability of a laser.

2.4.2 Mach-Zehnder Interferometer

A Mach-Zehnder interferometer uses two beam splitters and two mirrors to divide and subsequently recombine an optical field. Unlike the Michelson configuration, the two beams travel along separate optical paths before being recombined at a second beam splitter. The interference intensity at the output depends on the difference in optical path length between the two arms. This relationship is given by

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta L, \quad (2.10)$$

where ΔL is the optical path difference between the two arms. A change in the physical length or refractive index of either arm therefore produces a change in the relative phase of the two beams. The resulting change in output intensity can be used to detect small changes in optical path length. The Mach-Zehnder interferometer is therefore particularly useful for optical sensing and phase measurements, where changes in the optical path can be converted into measurable intensity variations.

2.5 Optical Atomic Clocks

An optical atomic clock uses the frequency of an atomic transition as a highly stable reference against which the frequency of a laser can be compared¹³. Atoms such as strontium (Sr) and ytterbium (Yb) possess extremely narrow optical transitions that can be used as frequency references. A highly stable laser is used to interrogate the atomic transition, with the laser frequency adjusted towards the resonance frequency of the atom. The difference between the laser frequency and the atomic transition can be represented as

$$\Delta f = f_{\text{laser}} - f_{\text{atom}}. \quad (2.11)$$

When the laser is resonant with the atomic transition,

$$f_{\text{laser}} \approx f_{\text{atom}}. \quad (2.12)$$

The response of the atoms to the laser can therefore be used to generate an error signal indicating whether the laser frequency is above or below the desired transition frequency¹⁴. Interferometric techniques can be used within such systems to measure and stabilise optical phase and path-length variation.

3 Free-Space Setup

This investigation aims to demonstrate an experimental approach to design, fabricate, and critically assess a low-cost and portable optical free-space setup. Several methods were deployed to develop the system and characterize its output, while comparing results with large-scale, widely used equipment.

3.1 Research Methodology - HS, BG

The methodology adopted for the free-space system centred on developing a practical experimental platform using a combination of in-house fabrication and selectively purchased optical components. Where possible, supporting hardware such as mounts, clamps, and structural elements was designed and manufactured specifically for the system, allowing the mechanical layout to be adapted to the required experiments while reducing dependence on expensive commercial optomechanical hardware. Components that could not reasonably be fabricated in-house, such as beam splitters, polarizers, lenses, and photodetectors, were selected and purchased according to the optical requirements of the experiments and the overall constraints of cost, size, and compatibility.

The methodology is therefore presented in two stages. First, the design and fabrication of the optical breadboard, mounts, and associated supporting components are described. This is followed by the assembly and configuration of the platform for the series of optical experiments used to characterise the laser source and evaluate the performance of the completed system.

3.1.1 Design and Fabrication of 3D Printed Optical Mounts - BG

A series of optical mounts were designed and printed to support the low-cost and compact construction and alignment of the system. The mounts were designed in Autodesk Fusion 360 and fabricated using Bambu X1C 3D printers, providing a low-cost alternative to traditional commercially available optical mounts. Both the optical breadboard and the laser mount were manufactured using the University of Bristol's machining workshop.

The optical breadboard features a 14 by 14 array of $M4 \times 0.7$ mm threaded holes for the optical mounts to be attached to, as well as four shock-absorbing rubber legs fitted to the bottom. The laser mount features a semi-cylindrical slot as seen in figure 4.1 with a low tolerance so that the laser can be fitted tightly. Clamps are used both to secure the laser, thereby increasing stability, and to hold a small thermistor in contact with the laser near its aperture for temperature measurement. Below the aperture of the laser is a small shelf used to house the Peltier module. As both the laser mount and the optical breadboard are aluminium, they can also act as heat sinks for the laser module.

The optical breadboard is housed within a box for portability and to reduce interference from surrounding light. The top of the box can be separated from the bottom, avoiding unnecessary shock to the system by removing the need to take the optical breadboard out of the enclosure. This is achieved by fitting 10 nuts into the top of the box

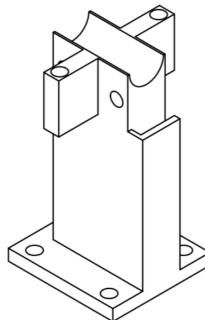


Figure 3.1: Sketch of Laser Holder

and 10 corresponding holes into the bottom for screws, creating a secure and fully portable free-space laser system measuring approximately $320 \times 320 \times 240$ mm when enclosed.

To allow the optical mounts to be adjusted along all axes, a post was designed with a hollow cylindrical tower for housing the cylindrical bases of the mounts. The tower includes a screw hole to allow adjustment of the height of the optical mounts and to keep the mounts stable. The base of the post allows positioning up to 16.5 mm away from any threaded hole in the breadboard, measured from the centre of the hole, by replacing a standard circular screw hole with an elongated slit in the base. If the post is required directly above a threaded hole in the breadboard, an additional hole is located at the bottom of the hollow tower.

A kinematic mirror mount was designed to allow fine adjustment of the beam direction while maintaining a low-cost construction. The mirror mount consists of four springs, three M4 screws, three M4 threaded nuts, a backplate, a mirror plate, and a mirror capsule. The four springs create four points of constant contact between the mirror plate and the backplate, while the three screws are used to adjust the angle of the mirror plate relative to the backplate. Two screws provide adjustment about the x - and y -axes, respectively, while the third screw is maintained at a constant position to provide a consistent pivot point for rotation of the mirror plate.

The mirror is press-fitted tightly into the mirror capsule to minimise vibrational disturbances that could cause unwanted deviations in the laser path. The mirror capsule is fitted with two small protrusions on each side, allowing it to be locked and unlocked from the mirror plate by twisting the capsule and thereby providing a secure and shock-resistant mounting arrangement. The backplate is equipped with a rod that fits into the post and is locked in place using the aforementioned screw in the hollow tower of the post.

The polarizer mount consists of two parts: the polarizer capsule and the polarizer capsule plate. The plate is designed with a rod that slots into the post and incorporates a cylindrical shell. The cylindrical shell features 1 mm diameter pegs positioned at 10° intervals to allow easy measurement of the polarization direction. The polarizer press-fits into the capsule, which includes three protrusions for locking the capsule into the plate. To allow rapid adjustment of the polarization, the capsule features an extended handle which, when paired with the 10° spacing of the pegs, creates a readable angular scale for the polarization direction.

Similarly, the lens mount consists of a lens plate and a lens capsule. The lens plate is fitted with a rod that attaches to the post and incorporates a cylindrical shell, while the lens capsule is fitted with two protrusions to allow it to be locked and unlocked from the lens plate. The lens capsule is curved, with an inner diameter approximately equal to that of the lens, allowing the lens to press-fit comfortably while resisting movement within the capsule. The curved edge of the lens capsule is sufficiently long to retain the lens securely while remaining short enough to avoid obstructing the laser beam as it passes through the lens.

The beam splitter mount features a slot at the top of a cylindrical shell, together with indentations around the inner surface of the shell. The top of the mount can be secured using a beam splitter lock which presses into the slot, holding the beam splitter in position and preventing it from falling out or deviating as a result of shock or vibration.

An iris was also designed to assist with laser alignment, particularly when configuring the laser system as a Mach-Zehnder interferometer (MZI). The iris consists of six overlapping blades, with both the upper and lower sides of each blade containing a protrusion for engagement with two separate plates. The first plate features six radial slits, into which the blade protrusions are inserted, converging towards a circular opening at the centre. The protrusions on the opposite sides of the blades are fitted into a rotating plate containing six corresponding holes and an extended handle.

When the rotating plate is turned using the handle, the ends of the blades are forced towards the central circular opening of the first plate, causing the blades to rotate while their opposite protrusions remain constrained within the rotating plate. This produces an adjustable aperture of progressively smaller diameter. Using two or more of these irises significantly simplifies optical alignment and reduces the time required to construct interferometric arrangements in which multiple beams must overlap, such as an MZI.

3.1.2 Basic Setup and Beam Alignment - HS

After multiple iterative prints, mounts and posts were fabricated to set the respective components into the optical breadboard. To assess stability of the fabricated mounts, a basic setup often called "walking the beam"¹⁵ is configured, a schematic of which is shown in Fig. (3.3).

A compact 30 cm × 30 cm × 0.8 cm optical breadboard was milled out of aluminum (for portability). Atop that, the posts were screwed in using M4 screws, a standard used throughout the entire fabrication process. To dampen vibrations, rubber mounts were screwed into each corner of the breadboard. This prevented any disturbances on the bench from being translated into the mounts.

Components were selected according to the requirements of the proposed portable free-space optical system. Where possible, commercially available components were combined with custom-manufactured mounts to reduce both cost and footprint.

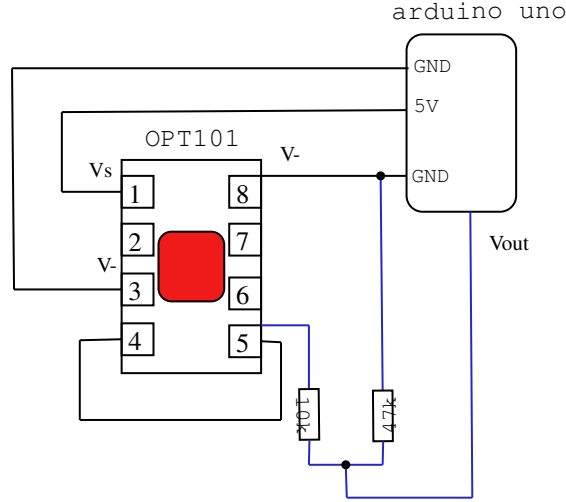


Figure 3.2: Photodiode (PD) circuit with respective ports labelled. The red region is the photosensitive part of the photodiode (PD). The PD is powered using an Arduino Uno. An external potential divider circuit using a 10 kΩ and a 47 kΩ resistor allows the output of the PD to be read by the Arduino. GND is ground. Vs is supplied voltage, V- is connected to GND.

Highly reflective mirrors were set at the 45° positions as shown in Fig. 3.3. A 650 nm laser with an internal collimating lens and a circular beam was used. For detection, a monolithic photodiode with an internal amplifier was powered using the 5 V output of an Arduino Uno R4. The photodiode output was passed through an external potential divider before being connected to an analogue input pin of the Arduino, allowing the corresponding voltage values to be recorded; the circuit is shown in Fig 3.2. The available degrees of freedom in the optical mounts, together with the iris, allowed fine adjustment of the beam path until the beam was fully aligned with the photodiode. The input voltage and output voltage both coming out of the Arduino module allows for more portability by removing requirement for an external power supply or voltmeter.

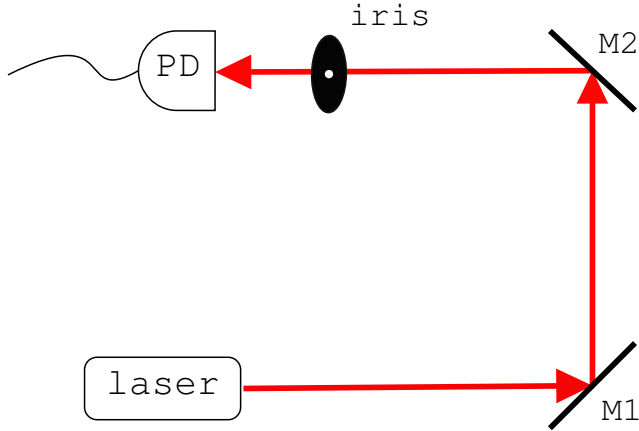
Beam adjusted using mirror mounts, and height of beam repetitively checked to make sure beam hit the center of all components. This ensured all parts were aligned.

The knife-edge technique was used to characterise the transverse spatial intensity distribution and beam size of the 650 nm diode laser. In this method, an opaque edge is translated incrementally across the laser beam while the transmitted optical power is recorded using a photodetector. As the knife progressively blocks the beam, the detected power decreases from its maximum value to the detector background level.

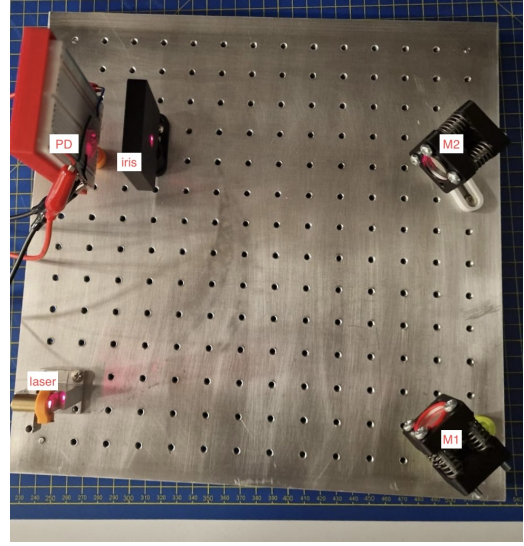
For an approximately Gaussian beam, the transverse intensity distribution can be expressed as

$$I(x) = I_0 \exp \left[-\frac{2(x-x_0)^2}{w^2} \right], \quad (3.1)$$

where (I_0) is the peak intensity, (x_0) is the beam centre, and (w) is the ($1/e^2$) beam radius. The photodetector



(a) Schematic of the beam-walking alignment procedure.



(b) Experimental beam-walking alignment setup.

Figure 3.3: Beam-walking alignment procedure: (a) schematic with photodiode (PD), mirrors (M1 and M2), iris, and laser labelled and (b) corresponding physical implementation.

does not directly measure the local intensity ($I(x)$), but instead measures the total optical power remaining on the unblocked side of the knife edge. The measured knife-edge response therefore corresponds to the integral of the Gaussian intensity distribution and takes the form of an error function,

$$P(x) = \frac{P_0}{2} \left[1 - \operatorname{erf} \left(\frac{\sqrt{2}(x - x_0)}{w} \right) \right], \quad (3.2)$$

where the negative sign corresponds to the direction in which the knife edge was translated across the beam.

3.1.3 Interferometry demonstrations - HS

To demonstrate that the setup could produce stable and verifiable results, interferometry experiments were performed and investigated.

Michelson Interferometer

The Michelson interferometer provided a useful demonstration of the capability of the miniaturized optical platform, as its operation requires precise beam splitting, reflection, recombination, and alignment within a compact footprint. Successfully producing interference in this configuration therefore provided a practical test of whether a standard free-space optical experiment could be reproduced using the reduced-scale, low-cost system.

The configuration of the Michelson interferometer was done as shown in Fig. 3.4. The beam emitted by the laser was first passed through lens L1 to increase the beam size before being incident on the beam splitter (BS). The beam splitter divided the incident light into two coherent optical paths, which were directed towards mirrors M1 and M2, respectively. After reflection from the mirrors, the two beams returned to the beam splitter, where they were recombined and directed towards the observation plane. Lens L2 was positioned after the beam splitter to further magnify the resulting interference pattern on the screen. L1 and L2 had focal lengths of 35 mm and 25.4 mm, respectively.

The optical geometry was kept compact while retaining sufficient propagation distance for alignment and observation of the interference pattern. The laser was positioned 6.8 cm from L1, while L1 was positioned a further 5.5 cm from the beam splitter. The distances from the beam splitter to mirrors M1 and M2 were 11.0 cm and

9.2 cm, respectively, giving a difference in arm length of 1.8 cm and a corresponding round-trip optical path difference of approximately 3.6 cm. Lens L2 was positioned 10.1 cm from the beam splitter along the output path. These dimensions allowed the complete interferometer to remain confined to the fabricated 30 cm $\times$ 30 cm optical breadboard.

As the Michelson configuration introduced a greater number of optical components and two separate propagation paths, additional care was required during alignment. The two arms were first aligned independently by temporarily blocking one optical path while adjusting the mirror in the other. This procedure was repeated for both M1 and M2 until the individual beams were directed towards the same region of the observation plane. Both arms were then unblocked, and fine angular adjustments were made to the mirrors to maximize the spatial and angular overlap of the returning beams. This iterative alignment ensured that the two wavefronts were sufficiently overlapped at recombination for a stable interference pattern to be observed.

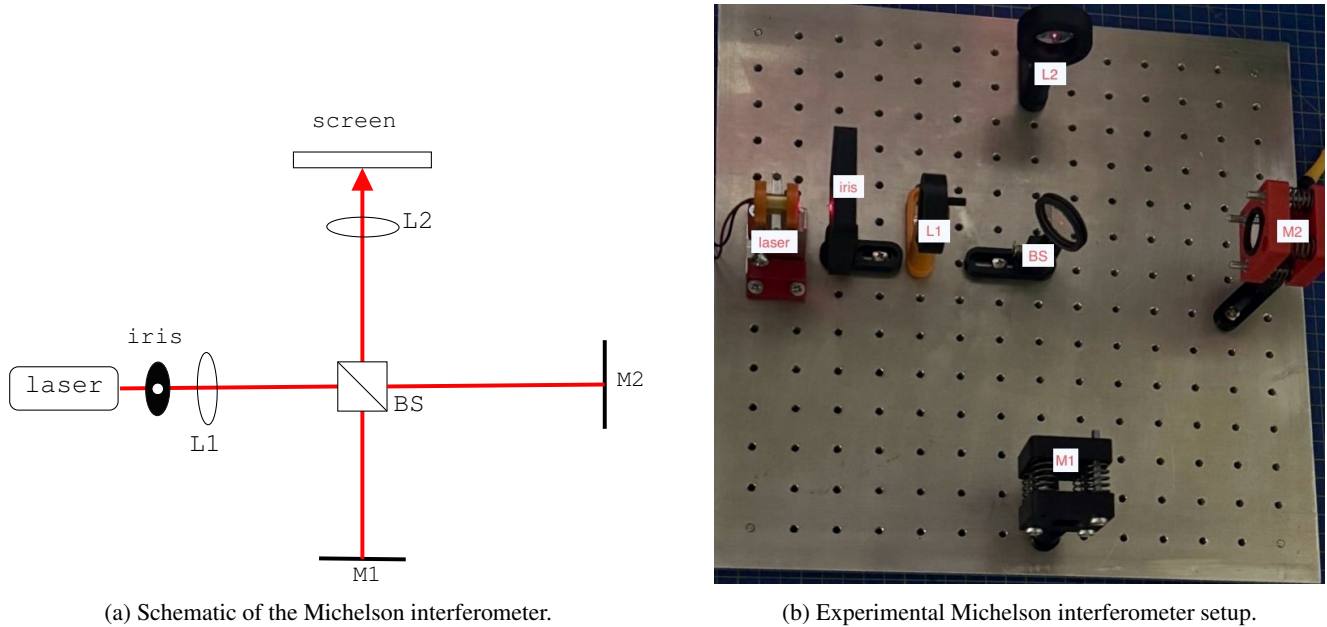


Figure 3.4: Michelson interferometer: (a) experimental schematic with beam splitter (BS), mirrors (M1 and M2), lenses (L1 and L2), laser and target screen. (b) corresponding physical implementation.

The sensitivity of the alignment arises from the dependence of beam position on small angular deviations in the optical components. For a propagation distance (L), an angular deviation $\Delta\theta$ produces an approximate transverse displacement

$$\Delta x \approx L\Delta\theta. \quad (3.3)$$

Furthermore, a rotation of a mirror by an angle θ changes the direction of the reflected beam by approximately 2θ . Consequently, even small angular errors in the mirrors could result in a significant displacement between the two returning beams, requiring fine adjustment during alignment.

In addition to spatial overlap, interference required the optical path difference between the two arms to remain within the coherence length of the laser and the polarization states of the returning beams to remain compatible. The phase difference between the two interfering beams can be expressed as in Eq. 2.7 where λ is the wavelength of the incident light and ΔL is the optical path difference between the two arms. Due to the double-pass geometry of the Michelson interferometer, a displacement (d) of one of the mirrors corresponds to an optical path change of

$$\Delta L = 2d. \quad (3.4)$$

Once satisfactory beam overlap had been achieved, the mounts and optical elements were kept fixed as far as possible, while external disturbances to the platform were minimized to reduce vibrations and prevent small

changes in the optical path length from destabilizing the interference pattern.

Mach-Zehnder Interferometer and Quantum Erasure

To further extend the interferometric capabilities of the miniaturized setup, a Mach–Zehnder interferometer (MZI) was constructed in the configuration shown in Fig. 3.5. The MZI provided a more complex demonstration of coherent interference than the Michelson configuration, while also forming the basis for investigating the wave–particle duality of light through subsequent polarization marking and which-path control. The initial configuration was therefore first aligned to produce stable interference before additional polarization elements were introduced.

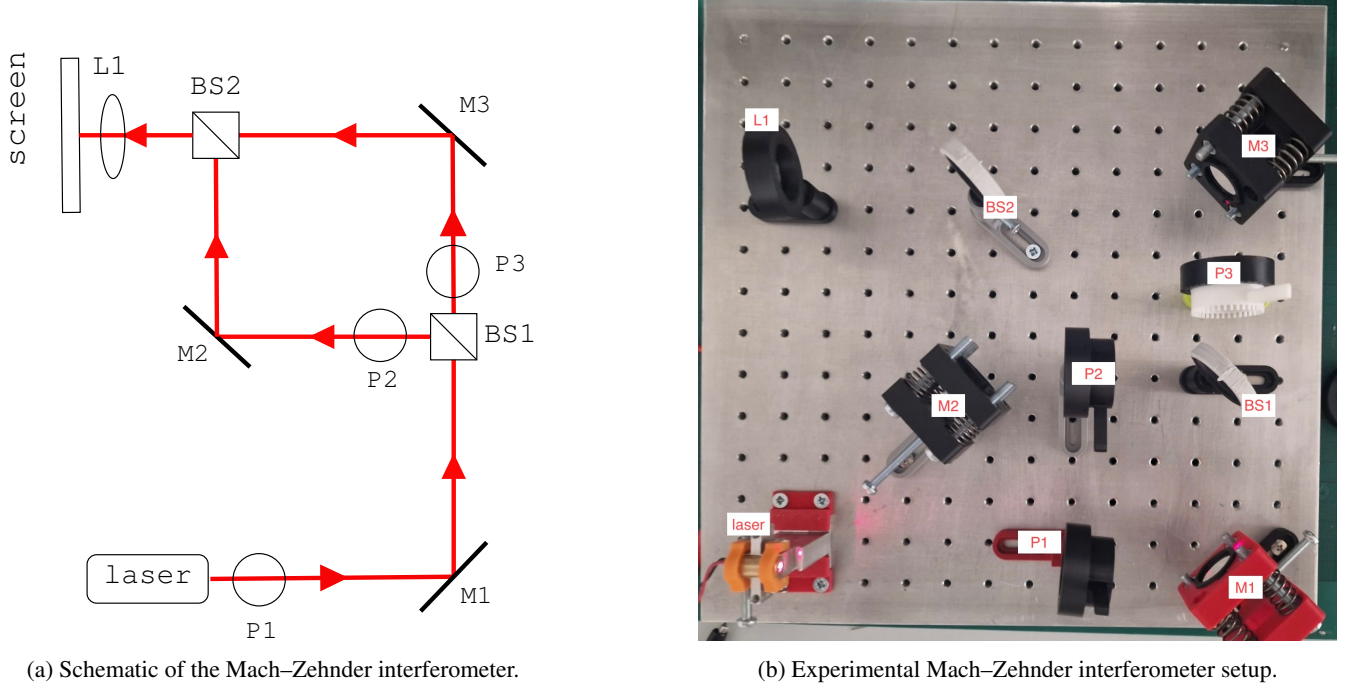


Figure 3.5: Mach–Zehnder interferometer: (a) experimental schematic with laser, polarizers (P1, P2, and P3), beam splitters (BS1 and BS2), mirrors (M1, M2 and M3), lens (L1), and a target screen, along with (b) corresponding physical implementation.

The incident light was first passed through a linear polarizer oriented at 45° , producing a beam with equal horizontal and vertical polarization components; the resulting polarization state can be represented as

$$\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle). \quad (3.5)$$

The polarised beam was then incident on the first non-polarising beam splitter, BS1, which divided the incident optical field into two spatially distinct paths. Since BS1 was non-polarising, both horizontal and vertical polarisation components could be either transmitted or reflected, and the beam splitter therefore acted primarily on the spatial degree of freedom rather than separating the two polarisation states.

A mirror was positioned in each of the two output paths from BS1 to redirect the beams towards a second non-polarising beam splitter, BS2. The two beams therefore propagated along separate optical paths before being recombined at BS2. Any difference between the optical path lengths introduced a relative phase difference between the two beams, given by Eq. 2.7. Provided that the two beams retained compatible polarisation states and were spatially overlapped at BS2, their coherent superposition produced interference at the two output ports.

For an ideal balanced MZI, the two output ports exhibit complementary interference behaviour. Thus, changes in the relative phase between the two arms redistribute the optical power between the two output ports while, ideally, conserving the total output intensity.

The increased number of components made alignment more demanding than for the Michelson interferometer. In addition to the reduction in optical power caused by transmission and reflection losses through the polarizer, beam splitters, mirrors, and other optical elements, small angular deviations could accumulate along the two independent beam paths and prevent satisfactory recombination at BS2. The trajectory of each beam was therefore mapped and aligned individually along well-defined straight optical paths before the two arms were recombined. Fine angular adjustment of the mirrors was then used to maximize the spatial overlap of the two beams at BS2 and subsequently at the observation plane.

To demonstrate quantum erasure, two additional polarizers, P2 and P3, were introduced into the two output paths of BS1. P2 was fixed at 0° , while the orientation of P3 was varied in increments of $\delta\theta = 10^\circ$. In the quantum description, the two spatial paths of the interferometer may be represented by the states $|1\rangle$ and $|2\rangle$. After BS1, the photon occupies a coherent superposition of the two paths, which may be written, neglecting relative phase factors, as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle) \otimes |\psi_{\text{pol}}\rangle. \quad (3.6)$$

The introduction of P2 and P3 correlates the path taken by the photon with its polarisation state. If the polarisation states in the two arms are denoted by $|p_2\rangle$ and $|p_3\rangle$, the state reaching the second beam splitter can be represented as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle|p_2\rangle + e^{i\Delta\phi}|2\rangle|p_3\rangle). \quad (3.7)$$

The visibility of the resulting interference therefore depends on the overlap between the two polarisation states, $\langle p_2|p_3\rangle$. When the two polarizers are similarly oriented, the polarisation states remain indistinguishable and the two paths can interfere. As the angular separation between P2 and P3 is increased, the polarisation states become increasingly distinguishable, providing increasing amounts of which-path information and consequently reducing the interference visibility. For ideal linear polarizers separated by an angle $\Delta\theta$, the polarisation-state overlap is

$$|\langle p_2|p_3\rangle| = |\cos(\Delta\theta)|, \quad (3.8)$$

and the interference visibility is therefore expected to decrease with increasing angular separation.

In the limiting case where the polarizers are orthogonal, for example P2 at 0° and P3 at 90° ,

$$\langle p_2|p_3\rangle = 0, \quad (3.9)$$

and the polarisation of the photon carries complete information about which interferometer arm it travelled through. The two alternatives are then distinguishable and interference is suppressed. In quantum-mechanical terms, a measurement capable of resolving these orthogonal polarisation states projects the photon onto one of the corresponding path-polarisation states, thereby destroying the coherent superposition between the two paths. The measurement therefore yields either $|1\rangle|H\rangle$ or $|2\rangle|V\rangle$, rather than retaining the original superposition of both possibilities. This loss of coherence is the origin of the disappearance of the interference pattern when complete which-path information is available.

3.1.4 Temperature Modelling - OF

Data of resistance vs temperature from the thermistor's data sheet (found in table 3.1¹⁶) were used to fit the Steinhart-Hart equation, an empirically derived equation which maps NTC thermistor resistance to temperature in Kelvin to high accuracy, shown below in equation 3.10¹⁷. Table 3.1 also includes the thermistor's alpha value at each data point which is defined below in equation 3.11¹⁸.

$$\frac{1}{T} = A + B\ln(R(T)) + C\ln(R(T))^3 \quad (3.10)$$

$$\alpha = \frac{1}{R(t)} \frac{dR(T)}{dT} \quad (3.11)$$

Least-squares polynomial regression produced the coefficient values for A, B and C shown in table 3.2, giving a maximum residual of 3×10^{-3} at 20° . α values from the Steinhart-Hart curve were calculated using the slope for the same temperatures as the manufacturer's data, residuals between those and the the data sheet's α values were used to validate the model. A maximum α residual of $0.045\%/^\circ\text{C}$ was measured.

Temperature ($^\circ\text{C}$)	$R(T)/R_{25}$	α ($\%/^\circ\text{C}$)
10	1.9900	-5.3113
15	1.5710	-5.1301
20	1.2490	-4.9574
25	1.0000	-4.7928
30	0.8057	-4.6358
35	0.6531	-4.4859
40	0.5327	-4.3428
45	0.4369	-4.2060
50	0.3603	-4.0753

Coefficient	Value
<i>A</i>	$1.125215708376 \times 10^{-3}$
<i>B</i>	$2.347142756703 \times 10^{-4}$
<i>C</i>	$8.577212639160 \times 10^{-8}$

Table 3.2: Steinhart–Hart Coefficients Determined by Regression

Table 3.1: Thermistor resistance ratio and corresponding temperature coefficient ($R_{25} = 10\text{k}\Omega$)¹⁶

The 14-bit ADC on the Arduino Uno Rev 4 minima with a reference voltage ($V_{REF;ADC}$) of 5V was used to sample the thermistor voltage and translate it to a temperature value by the following equations:

$$V_{ATO} = \frac{N_{ADC}}{2^{14}} V_{REF;ADC}, \quad (3.12)$$

Recovery of the voltage value at the MAX1978 ATO pin (V_{ATO}) from the raw bit measurement (N_{ADC}) at the Arduino ADC¹⁹.

$$V_{THERM} = \frac{4.53 - V_{ATO}}{4.03}, \quad (3.13)$$

Conversion of the Voltage at the ATO pin to voltage accross the thermistor, this can be derived from Figure 3 in the MAX1978 Evaluation Kit datasheet²⁰

$$R_{NTC} = R_{13} \frac{V_{THERM}}{V_{REF} - V_{THERM}}, \quad (3.14)$$

Current resistance value of thermistor, R_{13} which can then be plugged into the Steinhart-Hart equation (3.10) with calculated coefficients to obtain the temperature in Kelvin.²⁰. Equations 3.14, 3.13 are derived from values and circuit set ups in the MAX1978 data sheet.

3.2 Temperature control attempt - OF

3.2.1 Set-up

With the key constraints for this project being cost (£1050 budget with an aim of minimal usage), time (3 month timeline), safety regulations (maximum clearance for class II laser), size of the system (30cm x 30cm footprint) and portability - meant that only a few modalities on this list could be explored: temperature stabilisation and beam shaping.

Since the chosen laser source already includes a collimating lens, beam shaping was not a priority for the system. Initially, simplified frequency locking techniques which detect and correct the deviation of the laser's peak frequency (PDH locking being the gold standard²¹) were pursued but absence of wavelength-tunable lasers and lasers facilitating control of diode current for class II and below rendered this impossible. Use of external cavities for noise suppression, improved frequency stability and spectral line-width were also considered but a 10nm range for peak wavelength in the manufacturer's information and no access to equipment to properly measure the wavelength of the laser made this an impractical option.

The only possible variable to control the wavelength of the laser, therefore, was temperature and so an effort was made to design and install a thermoelectric control system to stabilise the temperature of the laser. Instead of using the laser beam to measure an error signal, as is done traditionally in laser locking, this system attempted to measure the temperature close to the laser diode and maintain a set value.

This system used a 5% tolerance 10k Ω NTC thermistor (model: TDK B57861S), 8.8mm² $\times$ 2.2mm Peltier module (model: SameSky CP2088-219) and an evaluation board based around an integrated circuit for thermoelectric control (model: MAX1978 Evaluation Kit).

Peltier modules consist of two ceramic faces sandwiching an array of alternating p-type and n-type 'legs' of semiconductors. When current is conducted through the array heat is transferred from one face to the other by the Peltier effect. Peltier modules are generally bi-directional, allowing for heating and cooling from both faces. The MAX1978 Evaluation Kit uses two PWM stages and filtering to send variable DC voltage and current to the peltier based on the voltage measured across the thermistor.

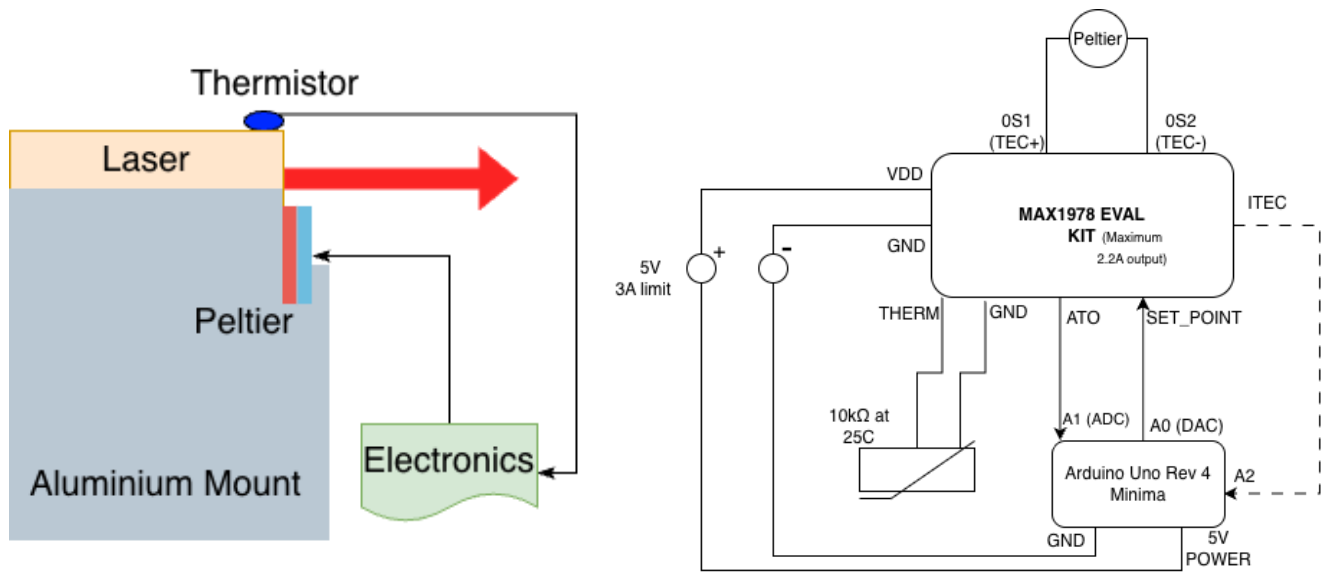
An aluminium mount was created to hold the laser with M4 screw holes for fastening to clamps and the aluminium base plate which housed the whole optical set up. In the laser mount a slot was machined for the Peltier on the front face, directly beneath the laser aperture. Clamps for the laser were 3D printed, one of which had a hole for the thermistor. The clamp was fastened over the top of the front of the laser module. Both the thermistor and the Peltier were applied with thermal paste to increase the thermal contact. The laser diode sits at the front of the laser module and so the placement of the thermo-electric components maximised proximity to the semiconductor source. The diagram in figure 3.6a gives visual aid to this description. The aluminium mount and base were designed to act as a large passive heatsink to conduct heat away from the Peltier, comprehensive thermal modelling was omitted due to the short timeline of this project and so it was assumed that this passive cooling would be sufficient. Measures were put in place to make this likely, namely that the temperature setpoint would be set only a couple of degrees above the temperature which the laser rose to after 15 minutes of operation. After implementing the TEC system the goal was to interrogate the effect of temperature stabilisation in interference patterns in order to get a qualitative figure of merit to assess the efficacy.

3.2.2 Diagnosis

An experiment was run to test the cooling and heating capabilities of the TEC, the thermistor measured the ambient temperature at 25.7°C (corresponding to a voltage of $V_{THERM} = 0.74V$). The set point voltage, $V_{SETPOINT}$, was set to 0.87V and then 0.712V, corresponding respectively to temperatures of 18.0°C (cooling case) and 27.5°C (heating case). The voltage at the ITEC pin, V_{ITEC} – which indicates the current supplied to the Peltier – was monitored while the TEC control loop was enabled.

In the heating case the thermistor quickly reached the set-point voltage but the temperature kept climbing with no sign of deceleration until the power supply was cut. The power supply was cut at $V_{THERM} = 0.65V$ to protect the laser as the upper limit of its operating range is 40°C. During the temperature climb The V_{ITEC} changed from positive to negative, indicating that the MAX1978 Evaluation Kit switched from heating to cooling mode despite the temperature continuing to climb.

In the cooling case V_{ITEC} remained negative throughout the experiment indicating proper cooling behaviour but the thermistor voltage diverged from $V_{SETPOINT}$ again, this time accelerating until $V_{THERM} = 0.65V$ was achieved and the power supply was cut off, again protecting the laser from overheating. These data clearly imply that the Peltier



(a) Diagram to show TEC set up with the laser, note that the clamps and baseplate are omitted for simplicity.

(b) Circuit diagram showing the pin layout for the TEC module (MAX1978 Eval Kit, thermistor and Peltier) and Arduino. The ATO pin transmits a voltage proportional to the thermistor resistance, the setpoint pin takes in a voltage corresponding to the set point temperature.

Figure 3.6: Schematics showing TEC control loop placement and wiring.

overheated.

3.3 Results - HS

The experimental analysis focused on directly measurable quantities obtained from the free-space system. Photodiode measurements were used to monitor detector voltage and its variation with time, while knife-edge beam profiling was used to determine the transverse beam profile, $1/e^2$ beam radius, and goodness of fit to a Gaussian distribution. Mechanical stability was quantified from changes in the detected signal following repeated external perturbations. Interferometric performance was then examined using Michelson and Mach-Zehnder configurations, with fringe spacing, fringe visibility, repeatability, and changes in interference contrast with polarizer orientation used to characterise the behaviour of the system.

3.3.1 Photodiode Measurements

The configuration shown in Figure 3.3 was first used to measure the response of the photodiode under different illumination conditions. The photodiode output provides a convenient measure of changes in incident optical power, allowing the response of the detection system to be compared under blocked, ambient-light, and direct laser illumination.

As shown in Figure 3.7, the photodiode produced a stable output of approximately 0.12 V when the detector was blocked. With the detector exposed to dim ambient illumination, the measured voltage increased to approximately 0.29 V. Upon direct illumination by the laser at measurement 11, the detector output increased sharply to approximately 4.02 V and remained stable for the subsequent measurements.

The clear separation between the blocked, ambient-light, and directly illuminated states demonstrates that the photodiode circuit can reliably distinguish changes in incident optical power. Since the measured output approaches the supply voltage of the detection circuit under direct laser illumination, the value should be interpreted primarily

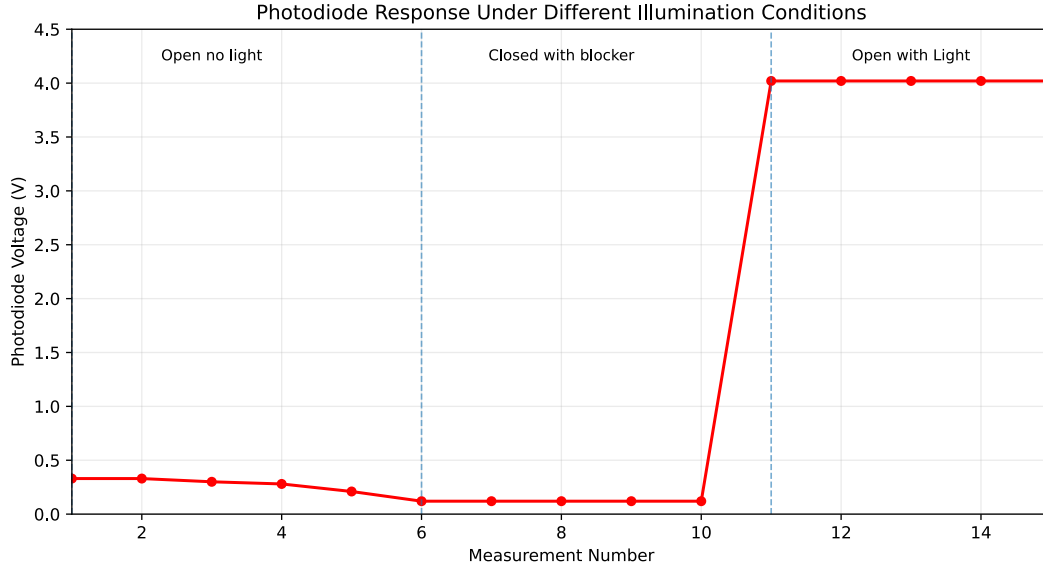
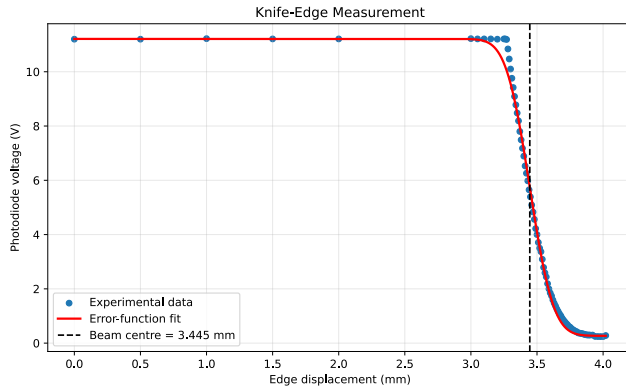


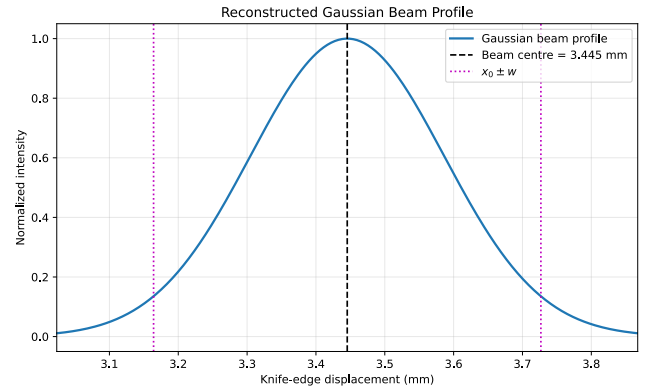
Figure 3.7: Photodiode voltage response under different illumination conditions.

as a relative detector response rather than an absolute measurement of optical intensity without further calibration of the photodiode and readout electronics.

The transverse spatial profile of the laser beam was subsequently characterised using the knife-edge technique. The experimental response is shown in Figure 3.8a, where the measured photodiode voltage is plotted as a function of knife-edge displacement together with an error-function fit. The characteristic sigmoidal response is consistent with the integrated transverse intensity distribution expected for a Gaussian beam.



(a) Knife-edge measurement.



(b) Reconstructed Gaussian beam profile.

Figure 3.8: Knife-edge measurement and corresponding reconstructed Gaussian transverse intensity profile.

The principal parameters obtained from the knife-edge analysis are summarised in Table 3.3.

The close agreement between the experimental measurements and the fitted error-function response, with $R^2 \approx 0.995$, indicates that the transverse beam profile is well described by a Gaussian distribution. The underlying intensity distribution can be reconstructed from the fitted knife-edge response because the derivative of the accumulated optical power is proportional to the local transverse intensity,

$$\frac{dP}{dx} \propto I(x). \quad (3.15)$$

The resulting reconstructed intensity distribution is shown in Figure 3.8b. For the Gaussian convention used in this analysis, the $1/e^2$ beam radius is related to the fitted standard deviation by

Table 3.3: Parameters obtained from the knife-edge beam-profile analysis.

Parameter	Symbol	Value
Beam centre	x_0	3.445 mm
Coefficient of determination	R^2	0.995
Gaussian standard deviation	σ	0.141 mm
$1/e^2$ beam radius	w	0.281 mm
$1/e^2$ beam diameter	$2w$	0.563 mm
Full width at half maximum	FWHM	0.331 mm

$$w = 2\sigma, \quad (3.16)$$

while the intensity full width at half maximum is given by

$$\text{FWHM} = w\sqrt{2\ln 2}. \quad (3.17)$$

The vertical dotted lines in Figure 3.8b indicate the positions $x_0 - w$ and $x_0 + w$, corresponding to the $1/e^2$ radius on either side of the fitted beam centre. The reconstructed profile therefore provides both a quantitative estimate of the transverse beam dimensions and experimental confirmation that the laser output is approximately Gaussian.

A single knife-edge scan characterises the beam at one longitudinal position and therefore does not independently determine the beam waist or divergence. If the measurement were repeated at several positions z along the direction of propagation, the evolution of the beam radius could be described by

$$w(z) = w_0 \sqrt{1 + \left(\frac{z - z_0}{z_R}\right)^2}, \quad (3.18)$$

where w_0 is the minimum beam waist, z_0 is its longitudinal position, and z_R is the Rayleigh range,

$$z_R = \frac{\pi w_0^2}{\lambda}. \quad (3.19)$$

Such measurements would additionally allow the beam divergence,

$$\theta \approx \frac{\lambda}{\pi w_0}, \quad (3.20)$$

to be determined and, for a non-ideal Gaussian beam, could be extended to determine the beam-quality factor M^2 .

3.3.2 Mechanical Perturbations

The mechanical stability of the miniaturised platform was assessed by monitoring the photodiode output while deliberately introducing external disturbances to the optical bench. A basic beam-walking configuration was used such that any significant displacement or angular misalignment of the optical components would alter the amount of light incident on the photodiode and consequently produce a change in the measured voltage.

During continuous data acquisition, a 2 kg mass was dropped from a height of approximately 10 cm onto the bench at intervals of approximately 10 s. The setup was also physically moved during the measurement to assess whether the alignment was preserved under handling. The resulting photodiode signal is shown in Figure 3.9.

The photodiode output remained approximately constant at 3.303 V throughout the measurement, with the observed variation remaining within approximately 1 mV. An uncertainty of ± 0.1 mV was assigned to the measured

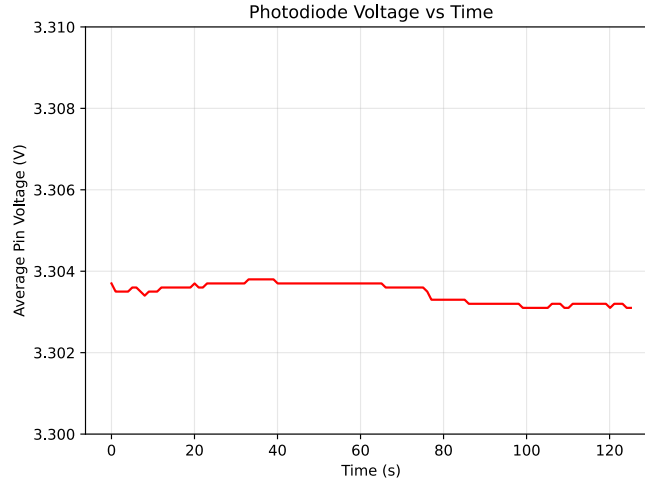


Figure 3.9: Photodiode voltage recorded while mechanical perturbations were applied to the optical platform. A 2 kg mass was dropped from approximately 10 cm onto the bench at approximately 10 s intervals, while the platform was also subjected to physical movement.

voltage stability. Relative to the mean photodiode output, the maximum observed variation therefore corresponds to approximately

$$\frac{1 \times 10^{-3}}{3.303} \times 100 \approx 0.03\%. \quad (3.21)$$

No substantial change in photodiode voltage was observed following either the repeated impacts or movement of the platform. As the beam propagated through two mirrors before reaching the photodiode, any significant angular or positional displacement of the mirrors or their mounts would have altered the beam trajectory and consequently the optical power incident on the detector. The measured stability therefore demonstrates that the optical mounts and overall beam alignment were preserved under the applied mechanical perturbations.

The optical breadboard was supported at each corner by shock-absorbing rubber mounts designed to mechanically isolate the system from external vibration and impact. The absence of any sustained variation greater than approximately 1 mV following repeated 2 kg impacts indicates that these rubber mounts effectively attenuated the transmitted mechanical shock, preventing significant disturbance of the optical alignment. Together with the stability of the mirror mounts, this demonstrates that the vibration-isolation design was effective in maintaining the optical configuration during external perturbation and physical handling. Although the measurement does not directly quantify the physical displacement of individual components, it provides a practical system-level assessment of mechanical stability and directly supports the objective of producing a mechanically robust and portable optical platform.

3.3.3 Interference Observations

Interference was successfully observed using both the Michelson and Mach–Zehnder interferometer configurations. The recorded fringe patterns were used to assess the behaviour of each interferometer and the response of the system to changes in optical path and polarization, and vibration.

Michelson Interferometer

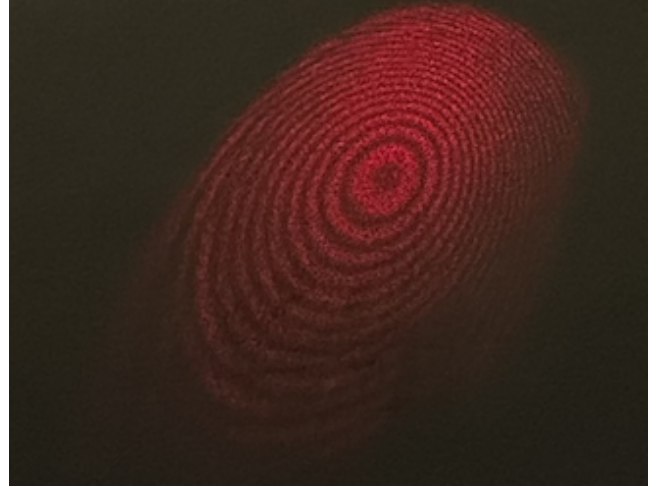
Before the interferometer was assembled, the spatial output of the laser was observed directly on a screen. The beam exhibited a weak concentric ring structure even in the absence of the Michelson interferometer, as shown in Figure 3.10a. This indicates that the laser output was not perfectly described by an ideal fundamental Gaussian

TEM₀₀ mode. Such spatial structure may arise from higher-order transverse-mode content, diffraction within the laser package, or weak parasitic interference between optical surfaces in the laser assembly^{22 23}.

Following alignment of the Michelson configuration, a substantially more regular system of concentric interference fringes was observed, as shown in Figure 3.10b.



(a) Spatial profile of the laser beam prior to interferometer alignment.



(b) Interference fringes observed using the Michelson interferometer.

Figure 3.10: Comparison between the unperturbed laser beam profile and the interference pattern produced by the Michelson interferometer.

The ring structure present in the unperturbed laser beam should therefore be distinguished from the interference pattern produced by the Michelson interferometer. While the former originates from the spatial characteristics of the laser output, the more regular fringes observed after recombination arise from the phase difference between the beams returning from the two interferometer arms. The circular fringe pattern results from a spatially varying optical path difference across the recombined wavefronts.

The clearly resolved fringes demonstrate that sufficient spatial overlap, coherence, and alignment were maintained within the miniaturised platform to produce stable two-beam interference. The characteristic spacing between adjacent fringes was measured to be approximately $630\text{ }\mu\text{m}$, providing a quantitative measure of the observed interference pattern.

Small perturbations to the optical path produced observable displacements of the fringe pattern, demonstrating the sensitivity of the interferometer to changes in relative optical path length.

Mach–Zehnder Interferometer

Following the Michelson experiment, the optical platform was reconfigured as a Mach–Zehnder interferometer (MZI). A clearly resolved linear interference pattern was obtained, demonstrating that the two independently propagating beams could be spatially overlapped and recombined while maintaining sufficient coherence to produce stable interference. The successful recovery of fringes following reconfiguration of the platform also demonstrates the repeatability of the fabricated mounting and alignment system.

To investigate the dependence of interference on polarisation, a linear polariser P_1 was placed before the first beamsplitter, while polarisers P_2 and P_3 were introduced into the two interferometer arms. For two linearly polarised optical fields, the detected intensity can be expressed as

$$I = I_2 + I_3 + 2\sqrt{I_2 I_3} \cos(\Delta\theta) \cos(\Delta\phi), \quad (3.22)$$

where I_2 and I_3 are the intensities of the two arms, $\Delta\theta$ is the relative angle between their polarisation states, and $\Delta\phi$ is the optical phase difference. The corresponding fringe visibility is therefore dependent on both the relative arm intensities and the overlap of their polarisation states,

$$V = \frac{2\sqrt{I_2 I_3}}{I_2 + I_3} |\cos(\Delta\theta)|. \quad (3.23)$$

For approximately equal arm intensities, this reduces to

$$V \approx |\cos(\Delta\theta)|, \quad (3.24)$$

such that maximum visibility is expected for parallel polarisation states, while orthogonal states ideally suppress the interference term.

Figure 3.11 shows representative interference patterns obtained under different polarisation conditions. A clearly defined fringe pattern was initially observed with $P_1 = 45^\circ$. As the polarisation conditions in the two arms were varied, the fringe contrast was progressively reduced, demonstrating the sensitivity of the MZI to the polarisation overlap of the recombined beams.

With $P_1 = 45^\circ$, $P_2 = 0^\circ$, and $P_3 = 90^\circ$, the two arm polarisations are nominally orthogonal. For an ideal system, the initial 45° polarisation provides non-zero projections onto both arm polarisers while P_2 and P_3 produce orthogonal output states. The polarisation overlap term therefore approaches zero, and complete suppression of the interference pattern would be expected. Experimentally, however, weak fringes remained visible at this nominal configuration, as shown in Figure 3.11b. The minimum fringe visibility instead occurred at a slightly displaced polariser angle.

This displacement indicates that the mechanical angular settings of the polarisers did not correspond exactly to the effective polarisation states at the point of recombination. The beams undergo several reflections and transmissions through the mirrors and beamsplitters, which may introduce different amplitude and phase responses for the *s*- and *p*-polarised components. Finite polariser extinction ratios and uncertainty in the angular zero positions may also contribute to the observed offset. Nevertheless, the progressive suppression of the interference pattern towards a minimum, followed by its recovery as the polariser angle was swept further, demonstrates the expected dependence of fringe visibility on polarisation overlap.

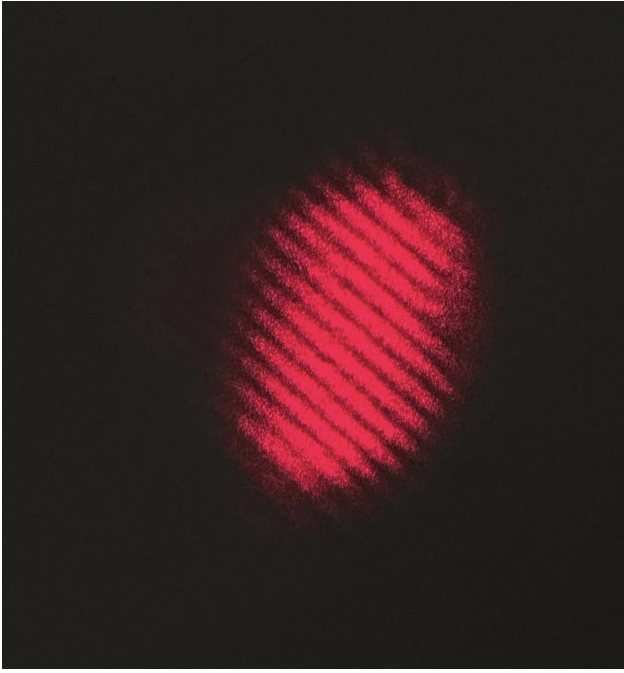
A second form of fringe suppression was observed with $P_1 = 0^\circ$ and $P_2 = 90^\circ$, with no P_3 present in the second arm. In this configuration, the intensity transmitted through P_2 is approximately described by Malus' law,

$$I_2 = I_{2,0} \cos^2(P_2 - P_1). \quad (3.25)$$

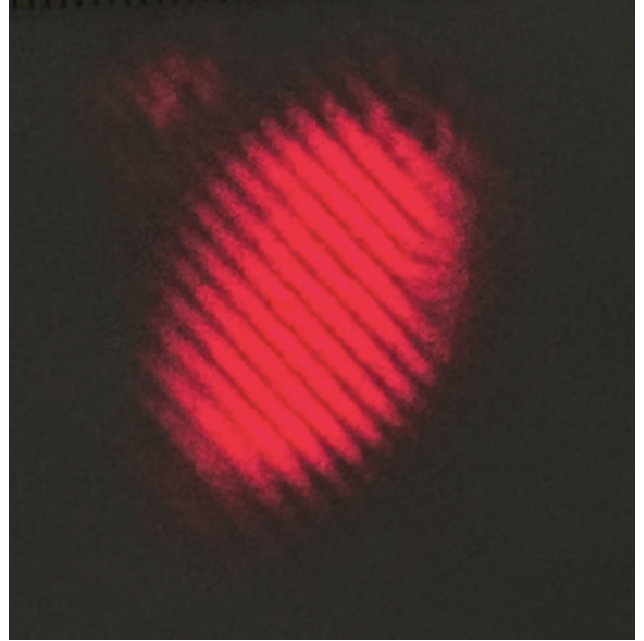
For an ideal system, $P_1 = 0^\circ$ and $P_2 = 90^\circ$ would therefore completely extinguish the optical field in the corresponding interferometer arm. Experimentally, however, the interference pattern was not completely eliminated and partial fringes remained visible, as shown in Figure 3.11c. This indicates that a finite optical field remained in the nominally extinguished arm, allowing weak two-beam interference to persist.

The residual transmission may arise from the finite extinction ratio and angular uncertainty of the polarisers, together with polarisation transformations introduced by the beamsplitters and mirrors. In particular, the polarisation state incident on P_2 may not have remained exactly aligned with the nominal 0° state established by P_1 . The observation of reduced rather than completely absent fringes is therefore consistent with strong, but incomplete, attenuation of one interferometer arm.

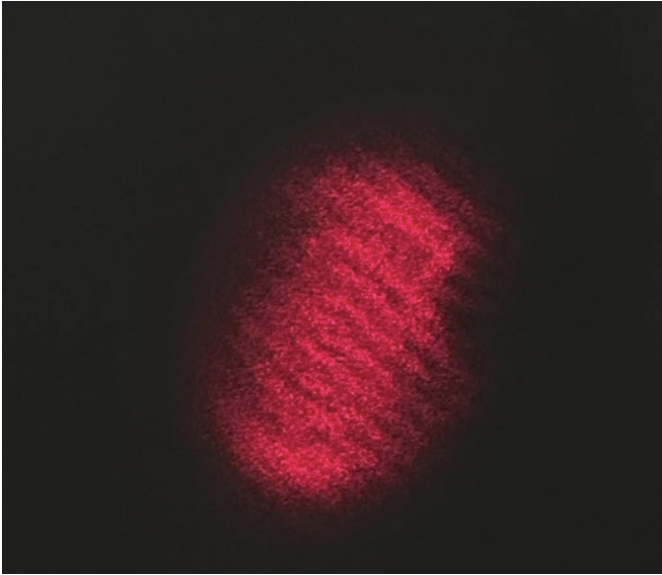
When P_3 was subsequently introduced and varied, further suppression of the interference pattern was observed. Near-minimum fringe visibility occurred for $P_1 = 0^\circ$, $P_2 = 90^\circ$, and $P_3 \approx 100^\circ$. Since the P_2 arm was already strongly attenuated, this minimum cannot be attributed solely to orthogonal polarisation states. Instead, it reflects the combined effects of arm-intensity imbalance and the polarisation overlap of the remaining optical fields at recombination.



(a) Initial MZI interference pattern with $P_1 = 45^\circ$.



(b) Nominally orthogonal configuration with $P_1 = 45^\circ$, $P_2 = 0^\circ$, and $P_3 = 90^\circ$.



(c) Partial fringe visibility for $P_1 = 0^\circ$, $P_2 = 90^\circ$, with no P_3 .



(d) Minimum observed fringe visibility during the polarisation sweep.

Figure 3.11: Variation of the Mach-Zehnder interference pattern under different polarisation conditions. Fringe visibility decreases as the effective polarisation overlap and relative arm intensities are altered.

A disappearance of the fringe pattern was additionally observed for $P_1 = 0^\circ$ and $P_2 = 45^\circ$. This behaviour is not predicted by the simplest ideal polariser model, since a 45° polariser would reduce the transmitted intensity but would not be expected to extinguish the corresponding optical field completely. The observation therefore suggests that additional polarisation transformations, arm-intensity imbalance, and alignment effects were present within the experimental MZI. This departure from the idealised model is considered further in the discussion.

A measurement taken of the intensity values from the photodiode at while sweeping through the different polarizations also demonstrates quantum erasure and Malus' law (Eq. 2.3), shown in A.2.

Overall, the MZI measurements demonstrate that the miniaturised platform was capable not only of producing stable two-path interference, but also of reproducing the expected qualitative dependence of fringe visibility on polarisation. The observed suppression and recovery of the interference pattern during polariser rotation provides a further functional validation of the platform beyond the Michelson interferometer experiment.

3.3.4 Temperature Characterisation - OF

The laser was powered up and left to run for 20 minutes while the Arduino Uno R4 Minima ADC sampled V_{THERM} at 100Hz. The ADC value was pushed to serial and the data was processed in Python. The ADC values were converted to temperature and at first an exponential decay curve was chosen to fit based on the visual trend of these 80 averaged points. The curvature of the single exponential decay term had a lower curvature than the 80 point's trend did and so a double exponential fit was trialled which gave a significantly better fit. The achieved fit is shown in the equations below and 3.12:

$$T(t) = T_\infty - A_{\text{fast}}e^{-t/\tau_{\text{fast}}} - A_{\text{slow}}e^{-t/\tau_{\text{slow}}} \quad (3.26)$$

$$T(t) = 23.010650 - 0.556185 \exp\left(-\frac{t}{0.495472}\right) - 0.546149 \exp\left(-\frac{t}{3.477582}\right) \quad (3.27)$$

$$[\text{RMSE} = 0.009^\circ\text{C}, \quad R^2 = 0.997]$$

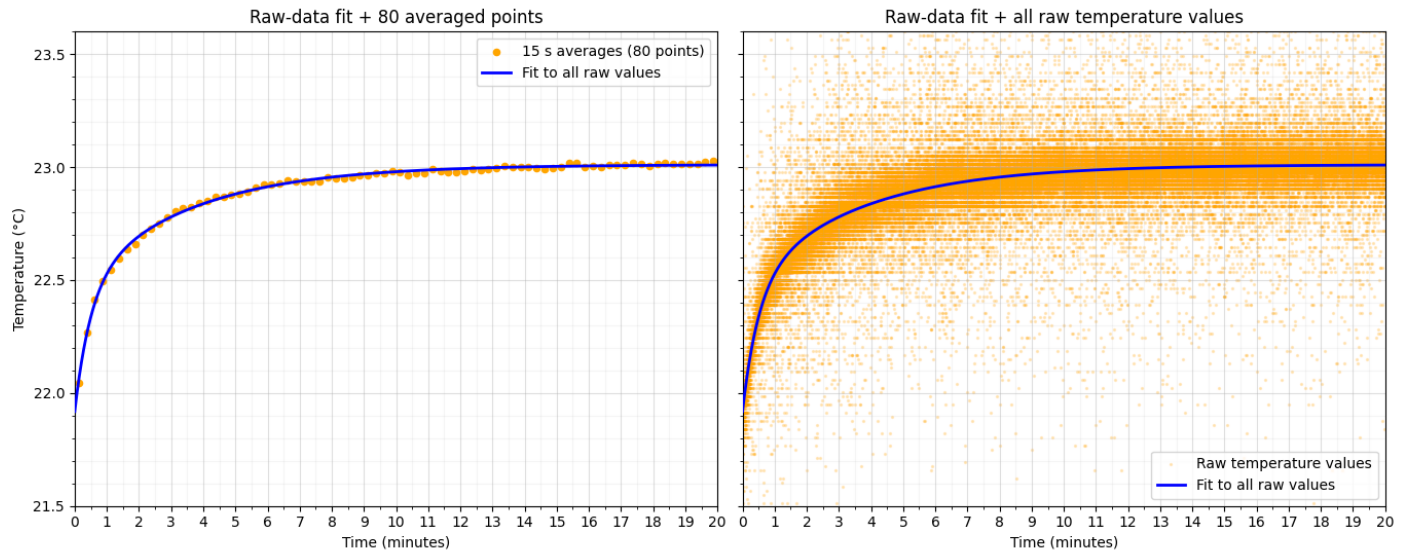


Figure 3.12: Side by side comparison of double-exponential temperature fit with all temperature samples and periodic average

Every 15 second period was averaged in order to omit the thermistor noise from the left-most plot and to give a clear view of the trend.

Within the trend of the averaged temperature points small, systemic deviations above and below the fit can be seen, More investigation is needed to determine whether this is due to the operation of the laser or if it is an artifact of the sensing apparatus. Without a feedback loop there should be no such behaviour as heat transfer should be passive

The fact that the convergent temperature seems to be stable to within $\pm 0.03^\circ\text{C}$ indicates that a temperature control system will likely not have a significant effect on a system using a class II laser and is superfluous. This is supported by the fact that interference patterns remained stable in time.

3.4 Discussion - HS

The results demonstrate that the fabricated platform achieved the principal objectives of the project: characterisation of the laser source, preservation of optical alignment under mechanical disturbance, and implementation of standard free-space interferometry. The system was also successfully reconfigured between Michelson and Mach–Zehnder geometries, demonstrating that the reduced footprint and low-cost mounting system did not prevent practical experimental use. The measurements nevertheless identified several limitations associated with detector calibration, polarisation control, and the alignment precision available from the fabricated components.

3.4.1 Laser Characterisation

The photodiode measurements confirmed that the detection circuit could clearly distinguish between blocked, ambient-light, and directly illuminated conditions. However, the illuminated output approached the supply voltage of the circuit, indicating operation close to saturation. The photodiode measurements are therefore most appropriately interpreted as relative changes in detected optical power rather than absolute intensity measurements. Future calibration against a known optical power meter, together with operation within the linear response range of the detector, would allow quantitative optical power measurements.

The knife-edge measurement provided a more quantitative assessment of the laser output. The close agreement between the experimental response and the error-function fit, with $R^2 \approx 0.995$, indicates that the overall transverse intensity envelope was well approximated by a Gaussian profile. A $1/e^2$ beam radius of approximately 0.281 mm was obtained at the measurement plane. This demonstrates that useful beam-profile information can be extracted using relatively simple components rather than requiring a commercial beam profiler.

The approximately Gaussian fitted envelope is not inconsistent with the weak ring structure observed in the directly projected beam. Fine spatial structure resulting from higher-order transverse components, diffraction, or parasitic interference may be superimposed on an otherwise approximately Gaussian intensity envelope. Since only one longitudinal position was measured, the value obtained represents $w(z)$ at that plane rather than necessarily the beam waist w_0 . Repeating the measurement along the propagation axis would permit determination of the beam waist, Rayleigh range, divergence, and M^2 factor.

3.4.2 Mechanical Stability

The mechanical perturbation experiment provided a system-level test of the stability of the fabricated platform. Despite repeated impacts from a 2 kg mass dropped from approximately 10 cm and physical movement of the system, the photodiode output remained stable to approximately 1 mV around 3.303 V, corresponding to a relative variation of approximately 0.03%.

Importantly, the laser beam propagated through two mounted mirrors before reaching the detector. Significant translation or angular displacement of either mirror would therefore have altered the beam trajectory and the optical power incident on the photodiode. The stability of the measured signal indicates that the fabricated mirror mounts retained alignment during the applied disturbances.

The result also supports the effectiveness of the rubber supports fitted beneath the optical breadboard. Mechanical impacts applied to the supporting bench did not produce a sustained change in the optical signal, indicating that the supports effectively attenuated the transmitted shock. This is particularly important for a portable system, where resistance to handling and external vibration is more relevant than for a permanently installed optical table.

3.4.3 Interferometric Performance

Successful generation of interference in both Michelson and Mach–Zehnder configurations provides the strongest functional validation of the platform. Stable interference requires sufficient spatial overlap, temporal coherence, and preservation of the relative optical phase. The observation of clearly resolved fringes therefore demonstrates that these requirements could be satisfied using the miniaturised breadboard and fabricated mounts.

In the Michelson configuration, regular circular fringes with a characteristic spacing of approximately $630\text{ }\mu\text{m}$ were obtained. Small changes in optical path produced visible displacement of the fringes, confirming the expected sensitivity of the interferometer to relative path length. The successful reconstruction of the platform as an MZI further demonstrates that the system was not dependent on a single permanently aligned configuration; interference could be recovered after the optical arrangement had been dismantled and rebuilt.

The MZI additionally demonstrated polarisation-dependent interference. Fringe visibility decreased as the overlap between the polarisation states in the two arms was reduced and subsequently recovered as the polariser angle was swept further. The visibility minimum did not occur exactly at the nominal orthogonal configuration. This is consistent with a practical optical system in which the mechanical polariser angles do not necessarily correspond exactly to the final polarisation states at recombination. Polarisation-dependent reflection and transmission through the beamsplitters and mirrors, finite polariser extinction ratios, angular calibration errors, and unequal arm intensities can all contribute to this displacement.

The partial fringes observed for $P_1 = 0^\circ$ and $P_2 = 90^\circ$ also indicate that the nominally extinguished arm retained some residual optical field. Conversely, the complete loss of visible interference for some other settings cannot be explained by polarisation overlap alone and is likely to reflect a combination of polarisation transformation, intensity imbalance, and changes in spatial overlap. These departures from the ideal model illustrate the additional effects present in a real compact interferometer while preserving the expected qualitative polarisation dependence.

3.4.4 Cost and Footprint Comparison

A key objective of the experimental system was to reduce the cost and physical footprint associated with conventional free-space optical experiments while retaining sufficient functionality for beam characterisation and interferometry. The total expenditure on purchased components was approximately £810.69, including the laser, mirrors, beam splitters, lenses, polarizers, photodiodes, electronic components, power supply, thermistor, vibration-isolating mounts, springs and associated hardware. This value is also a conservative estimate, as some purchased items, such as the data cable, were ultimately unnecessary and could be fabricated from readily available components. Additionally, several optomechanical mounts were designed and manufactured in-house rather than purchased commercially.

Commercial optical systems typically require significantly more supporting infrastructure. For example, a standard optical breadboard or vibration-isolated optical table may itself represent a substantial cost before the optical components, detectors and electronics are added. The system developed here therefore demonstrates that basic optical characterisation and interferometric experiments can be achieved at considerably lower overall cost.

The reduction in footprint is similarly significant. The system was constructed on an approximately $300\text{ mm} \times 300\text{ mm}$ platform, corresponding to an area of 0.09 m^2 . In comparison, a representative $600\text{ mm} \times 900\text{ mm}$ optical work surface occupies 0.54 m^2 , making the developed platform approximately 83% smaller in area. This substantially improves portability and allows the complete setup to be relocated more easily than a conventional optical-table arrangement.

The reduction in cost and size inevitably introduces compromises in rigidity, vibration isolation and long-term alignment stability. Nevertheless, successful laser beam profiling, photodiode measurements, and observation of Michelson and Mach–Zehnder interference demonstrate that sufficient stability and functionality were retained for the experiments considered. The system therefore provides a compact and comparatively low-cost alternative for pedagogical demonstrations and preliminary optical experiments.

3.4.5 Sources of Error and Experimental Limitations

Several sources of uncertainty affected the measurements. The photodiode response was not calibrated in absolute optical power and approached saturation under direct laser illumination. Voltage measurements were also limited by the resolution and noise of the Arduino-based acquisition system. Consequently, small variations in optical power may not have been fully resolved.

The knife-edge measurement was limited by the resolution and backlash of the translation stage, the finite sharpness and alignment of the knife edge, and manual positioning. Any tilt between the knife edge and the

transverse beam axis would introduce systematic error into the fitted beam width. Furthermore, measurements at only one longitudinal position prevent determination of the true beam waist and propagation parameters.

The interferometric measurements were particularly sensitive to angular alignment. Small rotations of the mirrors change the overlap and relative angle of the recombined beams, while changes in optical path length alter the phase. The observed fringe spacing therefore contains uncertainty associated with both image measurement and alignment. Air currents, temperature variations, and vibration may additionally modify the refractive index or optical path during measurement.

For the MZI, further uncertainty was introduced by the polarisers. Their mechanical angle markings provide only an approximate representation of the effective optical polarisation after propagation through the system. Beamsplitters and mirrors can produce different amplitude and phase responses for *s*- and *p*-polarised components, while finite extinction ratios prevent perfect suppression of an orthogonal component. Rotation or insertion of an optic can also cause slight beam steering, changing spatial overlap independently of polarisation. These effects provide likely explanations for the displacement of the experimentally observed visibility minimum from the ideal prediction.

Despite these sources of error, the principal conclusions are based on effects substantially larger than the corresponding measurement uncertainties: clear Gaussian-like beam behaviour, millivolt-level stability under mechanical perturbation, reproducible interference, and systematic changes in MZI fringe visibility with polarisation.

3.4.6 Thermal Control

The diagnostic voltage and current measurements indicate overheating of the Peltier and an inability to cool effectively. Since the MAX1978 is non-programmable and relies on analogue voltage signals there is no clear way to intervene in this hardware bug. In future rigorous thermal modelling of the TEC system should be performed to inform the decision of a heat sink. This peltier would certainly benefit from an active heat sink and would likely operate as intended with one, unfortunately a TEC system as complex as this was not in scope for this project.

Another option is to dynamically set $V_{SETPOINT}$ to the current value of V_{THERM} after the temperature has been over-shot in order to encourage the TEC power to drop to 0, enacting a graduated cooling rather than simply changing polarity. This regime would require close to zero power to be supplied to the Peltier for long enough for V_{THERM} to naturally equilibrate back to the original and desired value of $V_{SETPOINT}$, this is a challenging task because it means there must be extremely minimal lag between the dynamically allocated value of $V_{SETPOINT}$ and the current value of V_{THERM} . This method would therefore require involved embedded systems design and possibly extra hardware such that the Arduino's loop is not delayed by other peripheral tasks.

By temperature characterisation we have concluded that the TEC system is superfluous but we can conclude that a higher power laser system would likely require an active heat sink if it were to benefit from thermal control.

3.4.7 Future Work

Several extensions could improve both the quantitative accuracy and experimental capability of the platform. Photodiode calibration and operation within its linear response range would permit absolute optical-power measurements, while knife-edge scans at several longitudinal positions would allow w_0 , z_R , beam divergence, and M^2 to be determined. Fringe visibility could also be extracted quantitatively from the recorded MZI images using

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \quad (3.28)$$

allowing the measured polarisation dependence to be compared directly with the theoretical $|\cos(\Delta\theta)|$ response.

The Michelson interferometer could be extended from a qualitative demonstration into a quantitative metrology experiment. Translating one mirror by a known distance Δx and counting N passing fringes would allow the laser wavelength to be obtained from

$$\lambda = \frac{2\Delta x}{N}. \quad (3.29)$$

The same configuration could be used to measure small displacements and vibrations. At a conceptual level, this represents the same fundamental interferometric principle used in gravitational-wave detectors such as LIGO²⁴, where extremely small differential changes in optical path length are inferred from changes in the interference signal. The present platform could therefore provide a useful pedagogical analogue of a Michelson-based gravitational-wave interferometer, although its sensitivity would of course be many orders of magnitude lower.

The MZI could also be extended by introducing a fourth polariser, P_4 , after recombination to act as an analyser. Orthogonal polarisations in the two arms encode which-path information because each propagation path is associated with a distinguishable polarisation state. Selecting one of these polarisations with P_4 would retain this distinguishability, whereas projecting both onto a common basis, for example with the analyser at approximately 45° , would remove the polarisation distinction and allow interference to reappear. This would extend the present polarisation experiment into a demonstration of which-path marking and polarisation-based erasure²⁵.

With a coherent laser source, this would remain a classical-wave analogue of the underlying projection physics. Incorporating a single-photon source and photon-counting detectors would permit the same concept to be investigated in the quantum regime.

Overall, the results show that the fabricated platform provides a viable inexpensive and portable alternative for carrying out fundamental free-space optical experiments. Its performance does not replace the precision of a full-scale research optical table, but the demonstrated mechanical stability, reconfigurability, beam-characterisation capability, and interferometric performance indicate that a substantial reduction in cost and footprint can be achieved without losing the essential functionality required for pedagogical and preliminary research applications.

4 Photonic Integrated Circuit

The project aims to miniaturise laser stabilisation technologies. As such, a photonic integrated circuit (PIC) design is warranted.

The free-space setup operates at 650nm. On the advice of the Project Supervisor, the PIC was designed for a target wavelength of 679nm. The rationale was that this is the 3P_0 to 3S_1 repumping transition; the PIC would compare with other stabilisation research efforts²⁶.

4.1 Research Methodology - EW

Our research and development followed a hierarchical simulation methodology. Key operational requirements were established for a PIC-based stabilisation system to be used within a wider optoelectronic control circuit. Alternative PIC architectures were evaluated; the most suitable was selected. Design decisions and component choices were based on academic literature and PIC components were constrained by foundry design guidelines²⁷. Individual components — waveguides, bends, tapers and MMIs — were designed, characterised and optimised using FDE, FDTD and EME, then S-parameters were extracted. The S-parameters were combined in INTERCONNECT to construct and validate two AMZIs with different FSRs: these were then linked to form a whole-circuit model.

Results were analysed in the context of a likely optoelectronic circuit implementation. Wavelength-dependent INTERCONNECT outputs were converted into normalised error signals (E1 and E2) to form a dual-AMZI wavelength lookup. Collision analysis assessed wavelength and error signal uniqueness. A detector-noise model was developed. $\pm 3\sigma$ limits are a standard convention in statistical process control²⁸. We derived reliable estimates for minimum and maximum wavelength changes that can be reliably detected assuming a 3σ noise floor.

4.1.1 Key operational requirements - EW

The laser stabilisation application requires three operational characteristics from devices.²⁹ These are listed in Table 4.1 with the targets defined by this project.

Req. #	Title	Notes	Project target
1	Uniqueness	The setup maps each unique wavelength to one unique output across a specified operational wavelength interval.	Unique within $679nm \pm 0.5nm$
2	Sensitivity	Around the target wavelength, a small change in wavelength results in appreciable output change.	Detect frequencies changes of the order 1.5MHz.
3	Direction	The output inherently shows whether the input has increased or decreased.	Normalised error signal

Table 4.1: PIC requirements

A target collision-free range of 1nm centred on 679nm was chosen as a respectable locking range for an atomic-physics wavelength stabiliser. Literature doesn't define a sensitivity metric. However, work on dedicated 679/707-nm Sr repump lasers³⁰ gives the 679-nm transition linewidth as about 1.4 MHz³⁰. A reasonable benchmark is that a 1 MHz wavelength-equivalent frequency change yields *at least* a 3σ response at the output. A normalised error quantity can be calculated if a device has at least two outputs. The error measures "how far" from target the device is and the sign indicates "which side of the target" it lies.

4.1.2 Architecture choice - EW

A dual-AMZI architecture was selected. It's shown in Fig. 4.1 in the context of a wider optoelectronic control circuit. In the diagram, P1-4 are photodiodes; T1-4 are amplifiers converting photocurrent to voltage; **V** are fibre V grooves. The architecture has precedent^{31;32} and its operational characteristics meet the requirements:

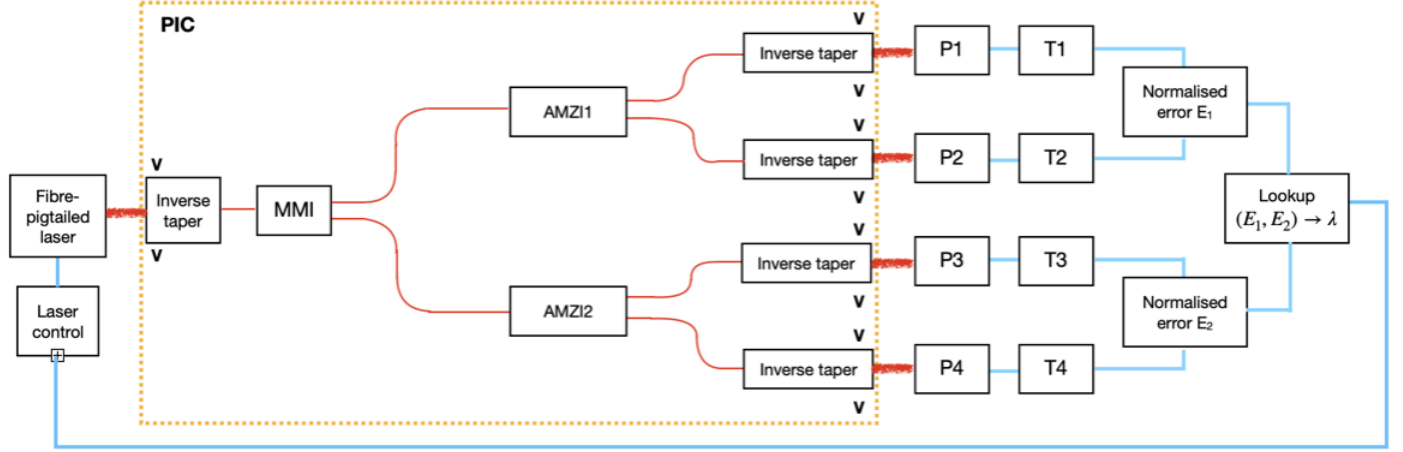


Figure 4.1: Optoelectronic wavelength discriminator circuit

Meets Req. #1: Uniqueness AMZIs have unequal FSRs and their outputs are processed into normalised error signals. Individual wavelengths map to (E_1, E_2) pairs. This offers coarse and fine wavelength discrimination and a broader operating range than a single AMZI alone.

Meets Req. #2: Sensitivity Error signal slope captures sensitivity $dE/d\lambda$. A shorter-FSR AMZI is at quadrature at 679nm; this ensures the error signal changes steeply with wavelength.

Meets Req. #3: Direction of wavelength change Error signals are calculated as follows:

$$E = \frac{P_1 - P_2}{P_1 + P_2}$$

Error signal values range -1..+1: sign inherently encodes direction of wavelength movement. Normalising in this way also removes common-mode noise.

This is a purely photonic, compact implementation. It doesn't require an atomic vapour cell or bulky external wavelength reference. The (E_1, E_2) error signal pairs could be precalibrated after fabrication: the PIC could be shipped ready for use.

4.1.3 Theoretical background - EW

The error values are cosine functions of wavelength, parameterised on the FSR and initial phase offset of AMZI1 and AMZI2.

$$E_{AMZI1}(\lambda) = \cos\left(\phi_1 + \frac{2\pi}{FSR_1}\lambda\right) \quad E_{AMZI2}(\lambda) = \cos\left(\phi_2 + \frac{2\pi}{FSR_2}\lambda\right) \quad (4.1)$$

For analysis purposes, the E_1, E_2 equations were expressed as:

$$E_{AMZI1}(\lambda) = \cos\left(\delta + \frac{2\pi}{FSR_1}\lambda\right) \quad E_{AMZI2}(\lambda) = \cos\left(\frac{2\pi}{FSR_2}\lambda\right) \quad (4.2)$$

where $\delta = |\phi_1 - \phi_2|$. Simultaneous measurement of AMZI outputs yields an error signal pair. Fig. 4.2 shows that two wavelengths can be indistinguishable to one AMZI ($E_1 = E_3$), but can be distinguished by reference to a second AMZI ($E_2 \neq E_4$). This is why a dual AMZI architecture can offer robust wavelength discrimination over a broader range than a single AMZI - and two outputs enable a normalized signed error output.

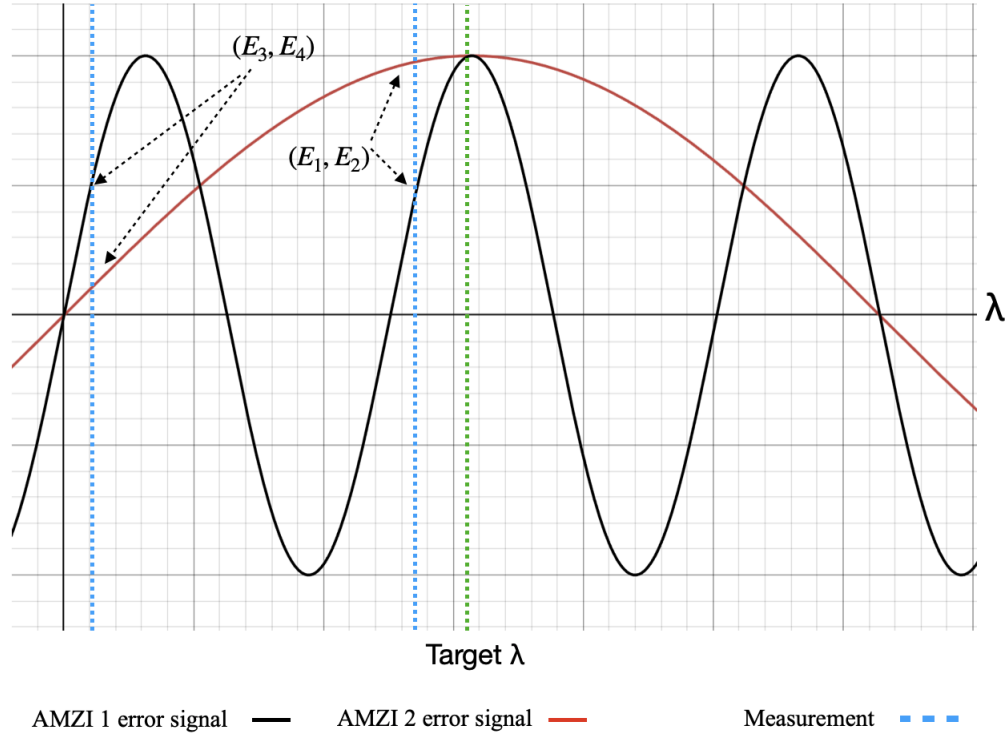


Figure 4.2: Mapping wavelength change to a unique output

Wavelength collision - EW

Collision occurs if two different wavelengths yield the same device output.

There are two cases. The wavelengths could be very similar; measurement noise makes them indistinguishable. Alternatively, since AMZIs are periodic, a combination of AMZIs will also have a period. So the wavelengths could differ substantially but if their wavelength distance matches this period, they will appear indistinguishable.

A detailed noise analysis is given later, resulting in a forecast of minimum detectable wavelength change. To discuss collision generally, consider parametrically plotting two cosine functions (an example is shown in Fig. 4.3). Any point where the curve crosses itself represents collision. Requirement #1 in Table 4.1 demands that, when AMZI outputs are plotted in this way, there must be no crossing for any two wavelengths in the range $679nm \pm 0.5nm$. This can be ascertained visually by plotting E_1 and E_2 around the $679nm$ point. It can also be determined analytically by line segment analysis.

4.2 Design - EW

Design parameters for AMZI1 could be selected from a basic assessment of circuit operation. Parameters for AMZI2 could be derived from there. The approach taken is discussed below.

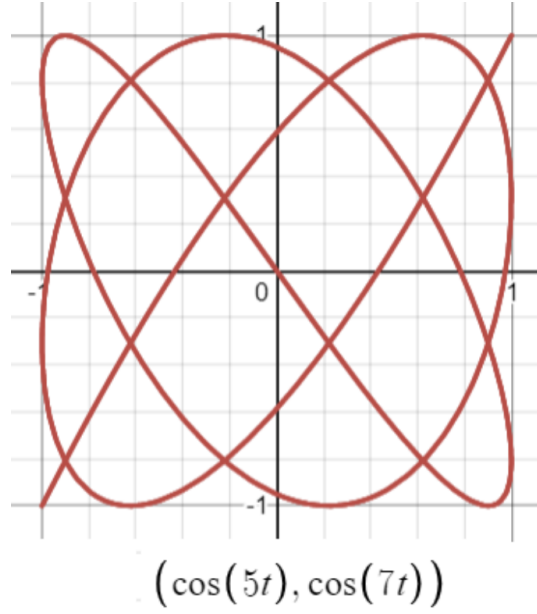


Figure 4.3: Parametric cosine plot example.³³

4.2.1 FSR choice - EW

Error signal slope measures wavelength sensitivity: high $dE/d\lambda$ means small $\Delta\lambda$ equates to pronounced ΔE . AMZI1, having a smaller FSR and hence generally steeper slope -provides sensitivity. A second, broader FSR for AMZI2 disambiguates wavelengths over a greater range - at the cost of its sensitivity, which is lost completely if $|dE/d\lambda|$ drops below a noise floor. Breadth vs sensitivity is the trade-off for FSR_2 . An FSR of 1.2nm was chosen for AMZI1; it offers decent error signal slope and it exceeds the wavelength range target. FSR_2 design began by aiming for FSR_2 between $5FSR_1$ and $6FSR_1$. Values in this range offer breadth and retain slope.

Collision between two wavelengths (λ_a, λ_b) is guaranteed when

$$|\lambda_a - \lambda_b| = \Delta\lambda = nFSR_1 = mFSR_2$$

for integers m, n . If m and n are simply related then $\Delta\lambda$ is minimised; $FSR_2 = 5.4, FSR_2 = 6$ were unsuitable. Some iteration was required to determine a value from the remaining range; a value of $FSR_2 = 5.3$ was selected (see section 4.4.8).

4.2.2 Initial phase offset - EW

For sensitivity around the target wavelength, AMZI1 slope must be at quadrature at 679nm. Its initial phase offset of $\phi_1 = 2.1rad$ reflected this.

Each AMZI arm carries an evolving optical field over it's optical path length. AMZI output is proportional to the phase difference between the fields in each arm.

$$P_{AMZI} \propto \cos(\phi_{AMZI})$$

But cosine is a symmetric function; different wavelengths *within the FSR* can evoke the same response:

$$\phi_{AMZI}(\lambda_b) = \pm\phi_{AMZI}(\lambda_a) + 2\pi n$$

for integer n . If this is met for both AMZIs simultaneously, collision occurs. This is a feature of trig functions and largely unavoidable, but adjusting the phase offset difference between the two AMZIs can maximise the collision-free range around a target wavelength.

Some iteration was required to determine an optimum relative phase difference. An Excel spreadsheet with 100 sample wavelength points over 678–681 nm was populated with error signal pairs generated from equations 4.1. A VBA macro treated each error signal pair as a point and formed 99 line segments between consecutive points. Every pair of non-adjacent segments was tested for intersection by solving simple simultaneous line equations. Where intersection occurred, the macro interpolated the corresponding wavelengths. The macro looped with fixed $\phi_1 = 2.1rad$ and ϕ_2 from $0..2\pi$ incrementing 0.025 rad each time. This is how the starting value of $\phi_2 = 3.14rad$ was selected.

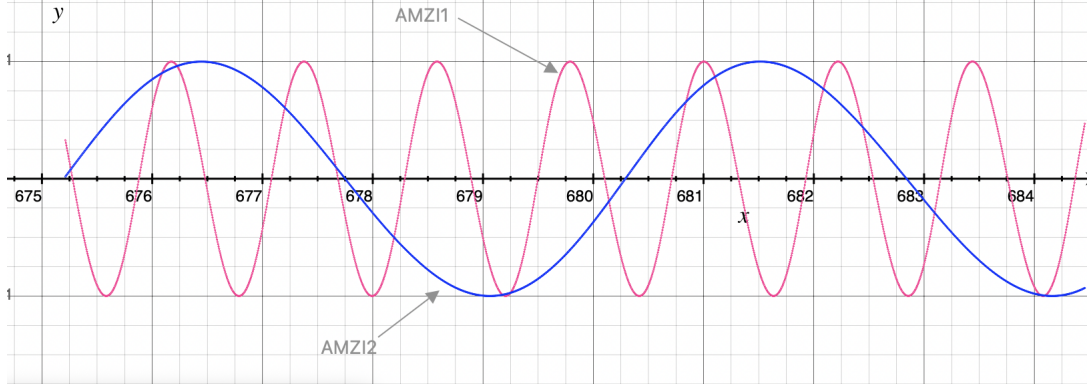


Figure 4.4: initial phase offsets

4.2.3 Circuit-level parameter summary - EW

The path length difference can be estimated from the *FSR*:

$$\Delta L \approx \frac{\lambda_0^2}{n_g FSR}$$

n_g can be obtained easily from FDE. Table 4.2 summarises design parameters that were used to satisfy the requirements (see Table 4.1 for a list of requirements). Note that the design *FSR* and ΔL are not exactly replicated in Interconnect - the realised response comes from the full optical network, not from an ideal formula.

AMZI	Design FSR (nm)	Interconnect FSR (nm)	ΔL (μm)	ϕ (rad)
1	1.2	1.21	163	2.10
2	5.3	5.34	28	3.14

Table 4.2: PIC design parameters

4.2.4 INTERCONNECT circuit and circuit simulation - EW

All elements of the circuit were included in the model as optical N-port parameter blocks. This included input/output tapers, waveguide straight and bend sections and MMIs.

Each simulation run sampled 2001 wavelength points. Broad sweeps used a range of 674–684nm; finer sweeps ran over 678–681nm. Optical input power was 2.5mW, assuming operation with a particular laser - details of the laser can be found in the Appendix A. Simulation data was processed by a range of custom Lumerical scripts developed for the project, by formulae in an Excel spreadsheet and detailed analysis was performed by a custom VBA macro developed for the project. Full details are in 4.4.

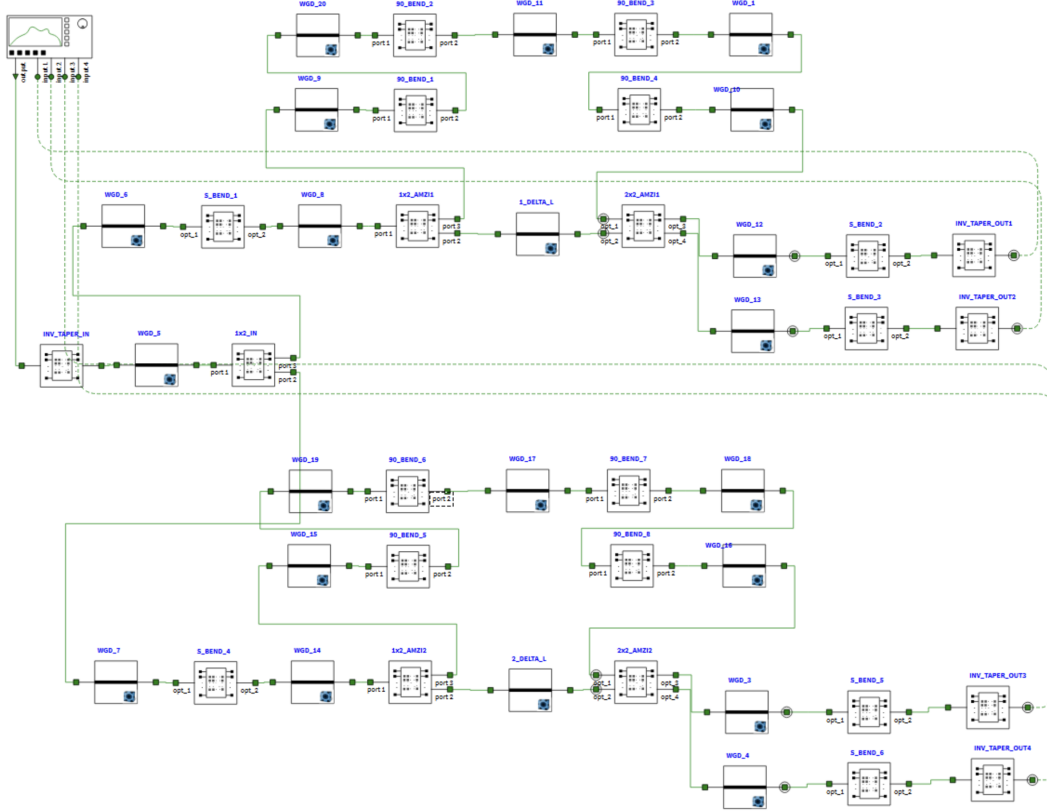


Figure 4.5: PIC circuit INTERCONNECT model.

4.2.5 Material stack - EW

Following foundry guidelines²⁷, components would be fabricated in Si_3N_4 with SiO_2 BOX and cladding.

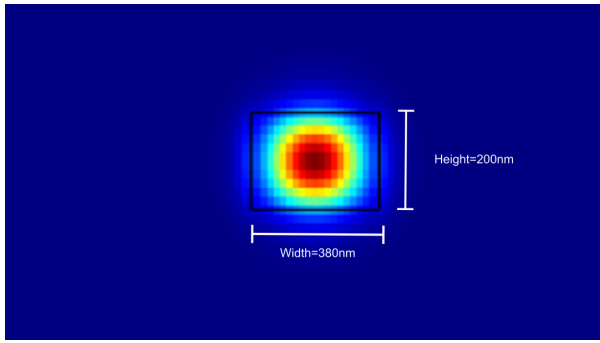
4.3 Components - PS

In Si Photonics technology, Silica (SiO_2) is typically used as the cladding, and the most popular materials used for waveguide cores and other passive components are Silicon Nitride (Si_3N_4) and Silicon Oxide (Silica), due to their broadband transparency and compatibility with planar wafer scale manufacturing in complementary metal-oxide-semiconductor (CMOS) or Si photonic foundries³⁴. SiN has a refractive index around 2, and it is deposited by plasma-enhanced chemical vapor deposition (PECVD) or low-pressure chemical vapor deposition (LPCVD). For current PIC design, Cornerstone PDK for SiN is followed with measurements constraints for height at 200 nm and width at 380 nm for waveguides and bends at visual range wavelength 679 nm. 2x2 couplers has separate design parameters for multi-mode convergence in MMI body.

4.3.1 Waveguide

Waveguides for the structure are simulated in FDTD for mode analysis and transmission using ports at 3 different lengths i.e. 10 μm , 12.6 μm and 72.1 μm . The mode profile for TE fundamental mode is common for most of the components, one shown below is extracted for a straight waveguide.

Transmission for all three waveguides is nearly the same and exists at $\approx 96\%$.



(a) Mode profile for waveguide 12.6 μm



(b) S-bend e-field

Figure 4.6: Waveguide and S-bend simulation results

4.3.2 Bend waveguide

90-bend and S-bend waveguides are chosen with a radius of 50 μm due to minimum bend radius constraint of 40 μm , for which simulated FDTD transmission stands at $\approx 99\%$ with a very low insertion loss of 0.043dB. E field is shown in Fig. 4.6b.

4.3.3 Edge coupler inverse taper

Edge couplers for light input and output from a Photonic Integrated Circuit is crucial as the actual fibre core way much bigger than the input waveguide. For highly efficient adiabatic edge coupler, the light propagates through the cladding material due to lack of mode confinement in small sized input inverse taper. Evanescent field then confined perfectly inside the taper after a certain length by following the high refractive indexed structure.

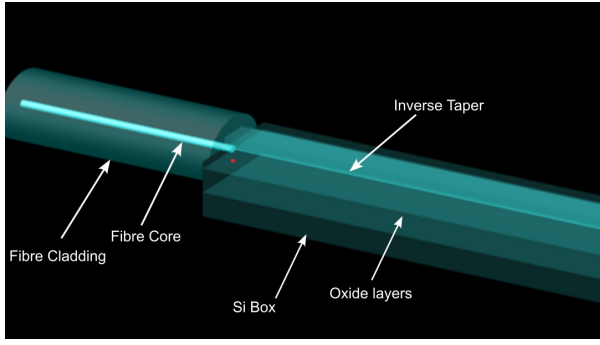
Major obstacle to achieving efficient fibre-to-chip edge coupling is the mode mismatch between the on-chip inverse taper and the fibre³⁴. We put the schematic and measurements of fibre cable with core diameter at 4 μm & cladding at 50 μm (only for simulation purpose) and an edge coupler, consisting of a 100 nm and 200 nm thick input waveguide, 400 μm long taper with narrow-side width of 100 nm and 380 nm at wider side, connected with another output waveguide as shown in Fig. 4.7a. The fibre measurements are taken from the reference of Wave Photonics SiN PDK for edge coupler at 674 nm with 630HP Single Mode Fibre of Mode Field Diameter ranges from 4.0 μm to 4.6 μm with NA=0.12 for Strontium ions.

This EME solver file is taken from Ansys-tutorial example file with manual changes were done for the material Refractive index and dimensions. Oxide and Box(SiO) layer on the top and bottom side of the taper structure are kept at 50 μm wide each to provide enough margin for field propagation in cladding.

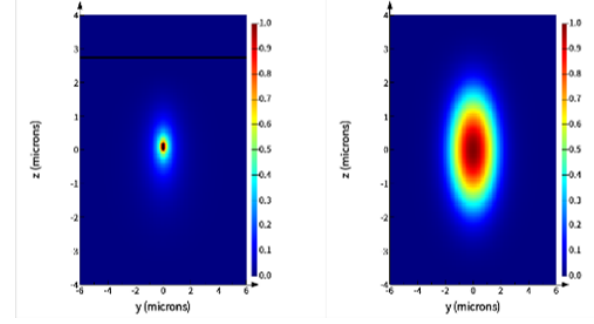
4.3.4 Mode overlapping

To get the mode overlapping efficiently, it is necessary to match the geometry in context of y & z-axis with the actual taper tip. For that, FDE solver was kept of same size to calculate modes for fibre and taper and after optimization for slight differences in axes of structures, we got 90.45% mode overlapping as shown in Fig. 4.7b. E field propagation through cladding into taper, e-field image was extracted after running EME solver propagation sweep, using frequency time monitor as in Fig. 4.7c. For better transmission of fundamental mode at the end tip of taper, total length of taper was kept at 400 μm . S-parameter sweep gave quite a high $T = |S_{21}|^2 = 0.96$ in Fig. 4.7d.

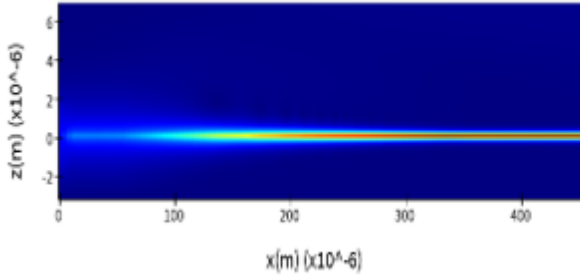
The Insertion loss for this edge coupler is 0.177dB which is acceptable. In initial attempts, we put a gaussian mode source with 2.0 μm beam radius for which Transmission was approximately 60% which is quite low for an edge coupling device.



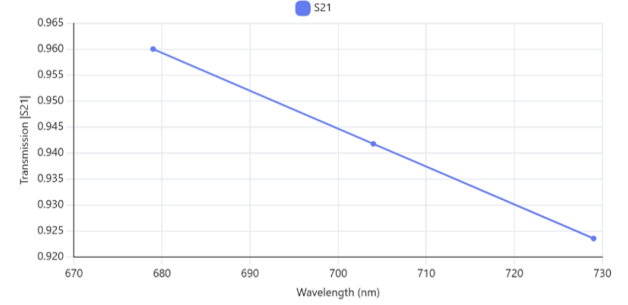
(a) Fibre-taper structure with oxide layers



(b) Mode coupling for fibre and taper in FDE solver



(c) E-field propagation from fibre core to inverse taper



(d) S-parameter T vs wavelength graph

Figure 4.7: Edge coupler simulation results

4.3.5 2x2 Multi-mode Interference coupler (MMI)

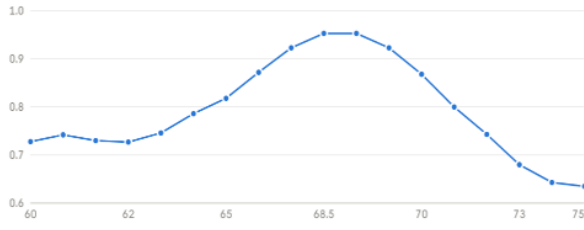
In silicon photonic waveguides, multimode interference (MMI) couplers hold a pivotal role in the realization of various advanced photonic devices, such as optical logic gates, thermo-optic switches, and arbitrary-ratio optical power splitters³⁵. The 2x2 MMI operates based on the self-imaging principle, where any input field is reconstructed periodically after a certain propagation distance (MMI body length). The length of the 2x2 MMI must be designed such that the input field from either port is reconstructed into two output fields at the end of the device in same phase³⁶. The width of the MMI is then set to $3.5 \mu\text{m}$ to avoid evanescent coupling of the output fields after the multimode region as per the PDK for 2x2 MMI. The mode separation distance is $0.5 \mu\text{m}$ between both ports of the 2x2 MMI with a taper length of $20 \mu\text{m}$. For MMI body length, taper lengths and pitch gap between taper for input and output, lengths sweeps were done in EME solver.

(a) Parameter sweeps

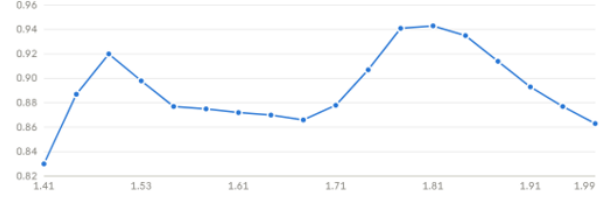
For a sufficient range of lengths for main body, tapers and pitch, we got different transmissions for wavelength centred at 679 nm through sweeps in EME solver as shown in Fig. 4.8a & Fig. 4.8b. Later, in FDTD simulations for these parameters, slightly different transmission with respective insertion loss results were found.

(b) Experimental simulations

From lengths parameters of EME solver, we simulated the structure with keeping same tapers measurements, for four different MMI body lengths i.e. 67, 68, 69 and $70 \mu\text{m}$ in FDTD to see the field propagation and multi-mode interference pattern in main MMI body. In initial phase of getting almost perfect 3dB coupler device, Mode sources were used instead of ports. As ports are highly accurate in terms of perfect mode transmission with very low scattering and insertion loss, low transmission was extracted on sources as inputs. For $70 \mu\text{m}$ length, using source, transmission curves are moderate and significant losses, while E-field propagation depicts proper



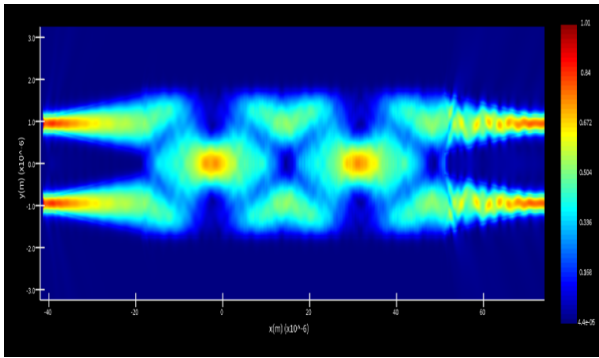
(a) MMI length vs Transmission



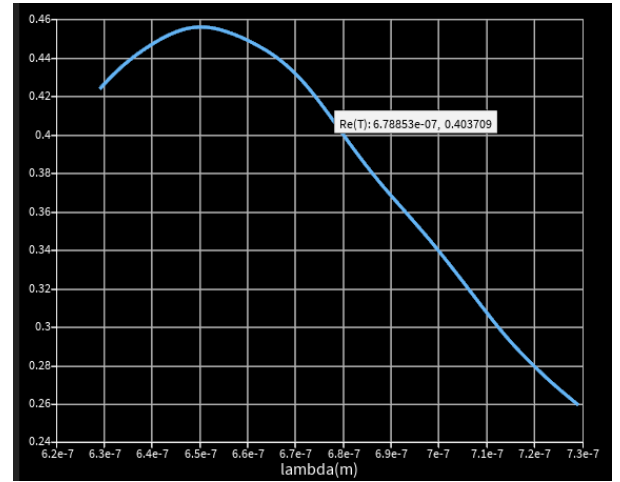
(b) Pitch length between tapers vs Transmission

Figure 4.8: 2x2 MMI parameter sweeps

interference patterns between multiple modes after mode convergence in MMI body as shown in Fig. 4.9a & Fig. 4.9b with need of more optimization.



(a) E-field in 2x2 MMI using mode sources at input ports



(b) Transmission vs wavelength at Port 3 and Port 4

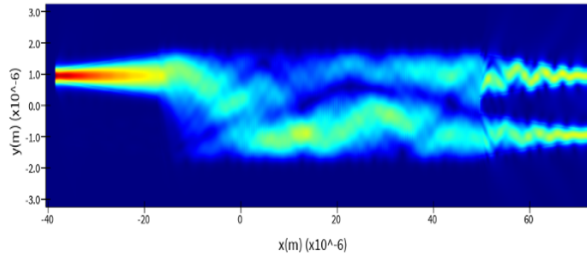
Figure 4.9: 2x2 MMI simulations using mode sources

For $67 \mu\text{m}$ to $70 \mu\text{m}$, using ports for the mode injection, a combined curve for Transmission vs wavelength over a span of 100 nm and centre of piece 2x2 MMI coupler at $67 \mu\text{m}$ with maximum transmission are shown in Fig. 4.10a & Fig. 4.10b.

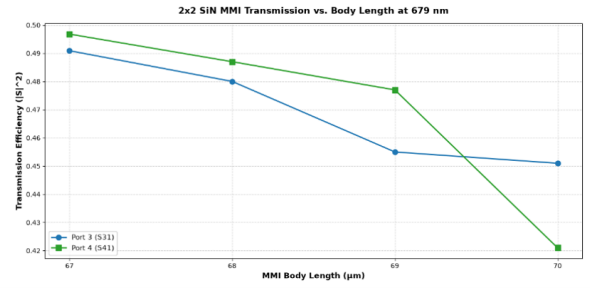
4.3.6 Phase difference analysis

The 2×2 MMI must provide two arms with inversely proportional field intensities, varying with the phase difference between both input ports because one output of the ideal 2×2 MMI contains a $\pi/2$ phase shift with respect to the other when fed through one input arm, and considering the reciprocal nature of the device³⁷. Through FDTD simulations, we tried to demonstrate the interferometric power routing capabilities of the 2x2 SiN MMI when both input ports are driven simultaneously with varying phase offsets. Because a restricted interference 2x2 MMI inherently relies on phase relations for self-imaging, altering the input phase delta drastically changes the output power distribution. In this study, two mode sources are applied to the inputs of the structure, while one's phase is swept from 0 to $\pi/4$ & $\pi/2$ radians and in other case where one's phase is kept at $\pi/2$ while putting other with a difference of π i.e. $\pi/2$ ³⁶. The normalized power transmission through each output arm is plotted for these sweeps and the lateral field for 3 cases as in Fig. 4.11.

The E field for 45° phase difference represents an intermediate interference state. The symmetric self-imaging condition is perturbed, resulting in an asymmetric mode beating pattern. The diagonal interference trajectory steers most of the optical power toward the top output Port 3, while the bottom waveguide receives a diminished, but

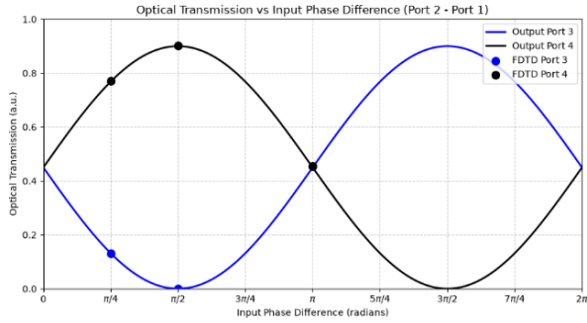


(a) E-field for 67 μm 2x2 MMI

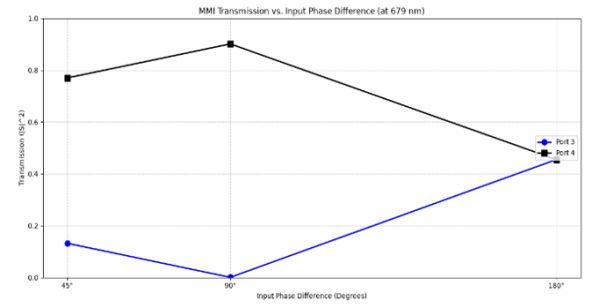


(b) Combined graph for Port 3 and Port 4 for all 4 MMI body lengths

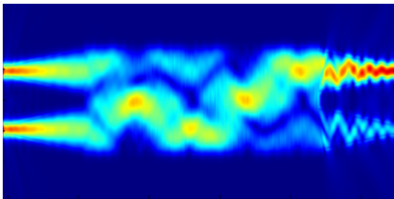
Figure 4.10: 2x2 MMI simulation results using ports



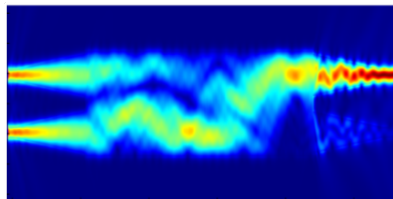
(a) Reference 2x2 MMI transmission in each output port vs phase difference between each input port



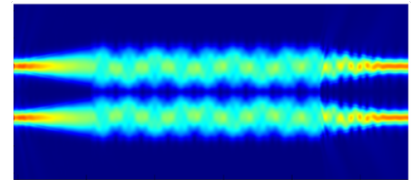
(b) Transmission on port3 and port4 along the phase difference



(c) $\pi/4$ phase difference



(d) $\pi/2$ phase difference



(e) π phase difference

Figure 4.11: Reference 2x2 MMI transmission and mode cross sections for varying phase difference between each input port. Propagation occurs from left to right.

non-zero, fraction of the field. In this case, the 2x2 MMI starting to act as a switching devices behaviour with Port 4 transmission is 13.2% and Port 3 transmission is 77.0% with total throughput is roughly 90%.

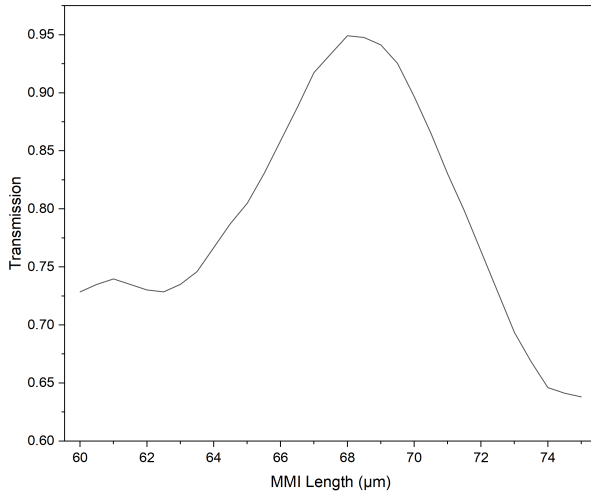
Unlike the previous case, in second case of phase difference of $\pi/2$, device acts like an optical switch by focusing most of input power into one output Port 3. For another trial to receive all the power out at Port 4, switching of current phase difference seems the appropriate way. The power distribution from the monitor at output ports shows Port 4 hits an almost perfect null at 0.1%. Port 3 reaches peak constructive interference at 90.1%. This is the cross-state condition for this phase difference setting.

For 180° phase difference one source to $\pi/2$ and the other to $\pi/2$, this creates a purely anti-symmetric excitation. The internal multi-mode interference pattern exhibits a strict zero-intensity null along the exact centre longitudinal axis ($y=0$) due to complete destructive interference at the midpoint. The field subsequently reconstructs symmetrically at the exit boundary, splitting the optical power equally between both output ports, like a 0° in-phase condition but with an inverted internal mode profile. This could be a mirror pattern of pure anti-symmetric pattern of 180° but with one source at 0° and other with a phase difference of 180° . For this phase setting of input sources, Port 3 and Port 4 transmissions reach approximately 45.5% each.

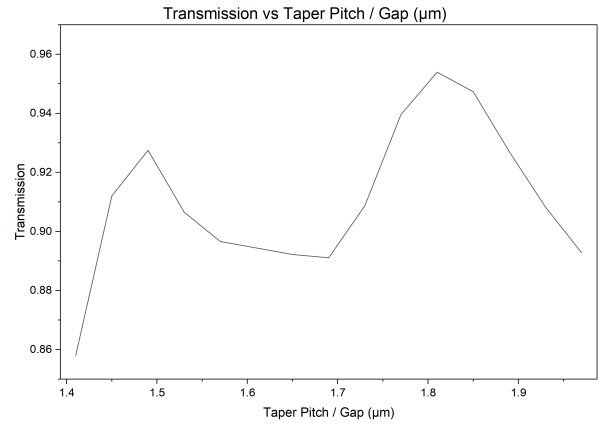
4.3.7 1x2 Multi-Mode Interferometer (Initial Splitter) - AL

While a standard Y-branch splitter is a common alternative, 1x2 MMI fundamentally relies on an infinitely sharp junction vertex to divide the optical mode symmetrically. The 1x2 MMI is fed by a standard 380 nm wide single-mode waveguide and inverse taper for single to multi mode conversion.

To facilitate an adiabatic transition into the multimode region and minimize scattering loss, the input waveguide gradually expands to a width of $1.4\ \mu\text{m}$ over a $20\ \mu\text{m}$ long input taper. MMI body length for maximum in phase interference is set at $17.5\ \mu\text{m}$ and later optimized at $16.75\ \mu\text{m}$ using length sweeps in EME solver. The core structure, lengths and pitch(taper gap) sweep results are shown in Fig. 4.12.



(a) 1x2 MMI body length sweep vs Transmission



(b) I/O taper's pitch gap sweep vs Transmission at 679 nm

Figure 4.12: 1x2 MMI parameter sweeps

Experimental simulations

For both $16.75\ \mu\text{m}$ and $17.5\ \mu\text{m}$ lengths, we ran FDTD simulations to observe the E-field patterns and overall transmission and phase at output ports. From Transmissions at Port 2 and Port 3 for both designs we got almost 50:50 output split which is near the target. On comparing the results, both designs were extremely efficient at 679 nm centred wavelength with a margin of 10 nm with transmission of 49.74% and 48.76% and zero degrees of phase difference. Lateral e-field propagation and transmissions at output ports for both structures are shown in Fig. 4.13.

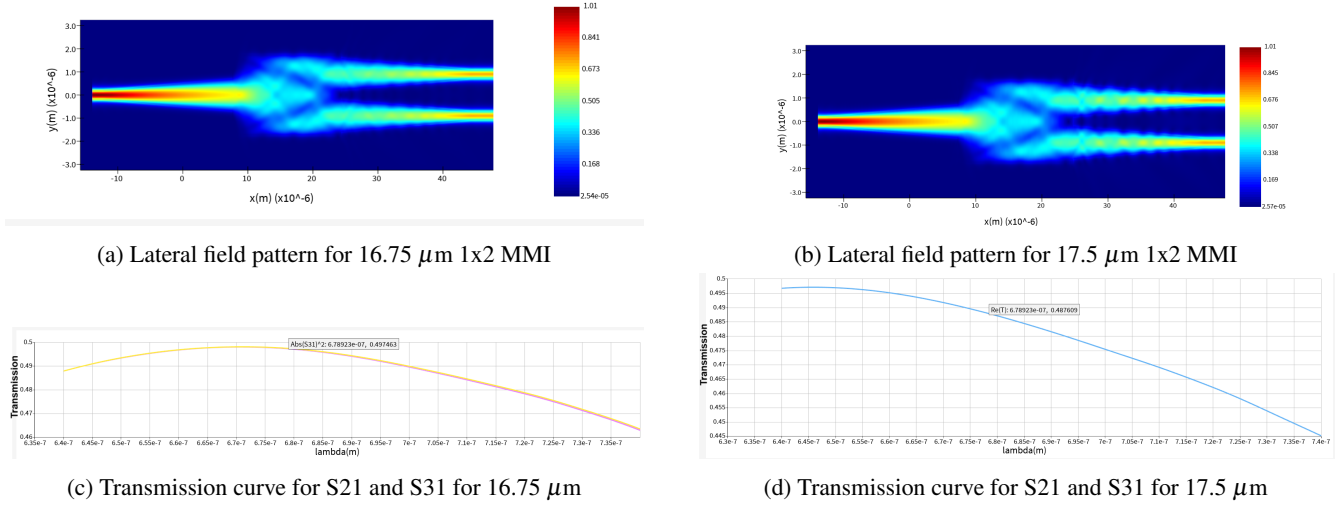


Figure 4.13: FDTD simulation results for the 1x2 MMI

4.3.8 Fabrication-Implementation Details and Material Choices - AL

The photonic chips geometry entirely depend on the operating wavelength. This circuit uses 679 nm red light to successfully control strontium atoms and secure them from getting lost. Because regular standard silicon absorbs this light, we used a silicon nitride platform which is completely transparent to visible light and also it offers low power loss, high thermal stability. The circuit was specifically designed to match the needs of the manufacturing process, that's why the circuit followed the cornerstone PDK (process design kit) 200 nm SiN Platform. To make sure the light travels through the circuit in the correct way, and the components were designed using KL layout and GDS factory (Fig. 4.14)

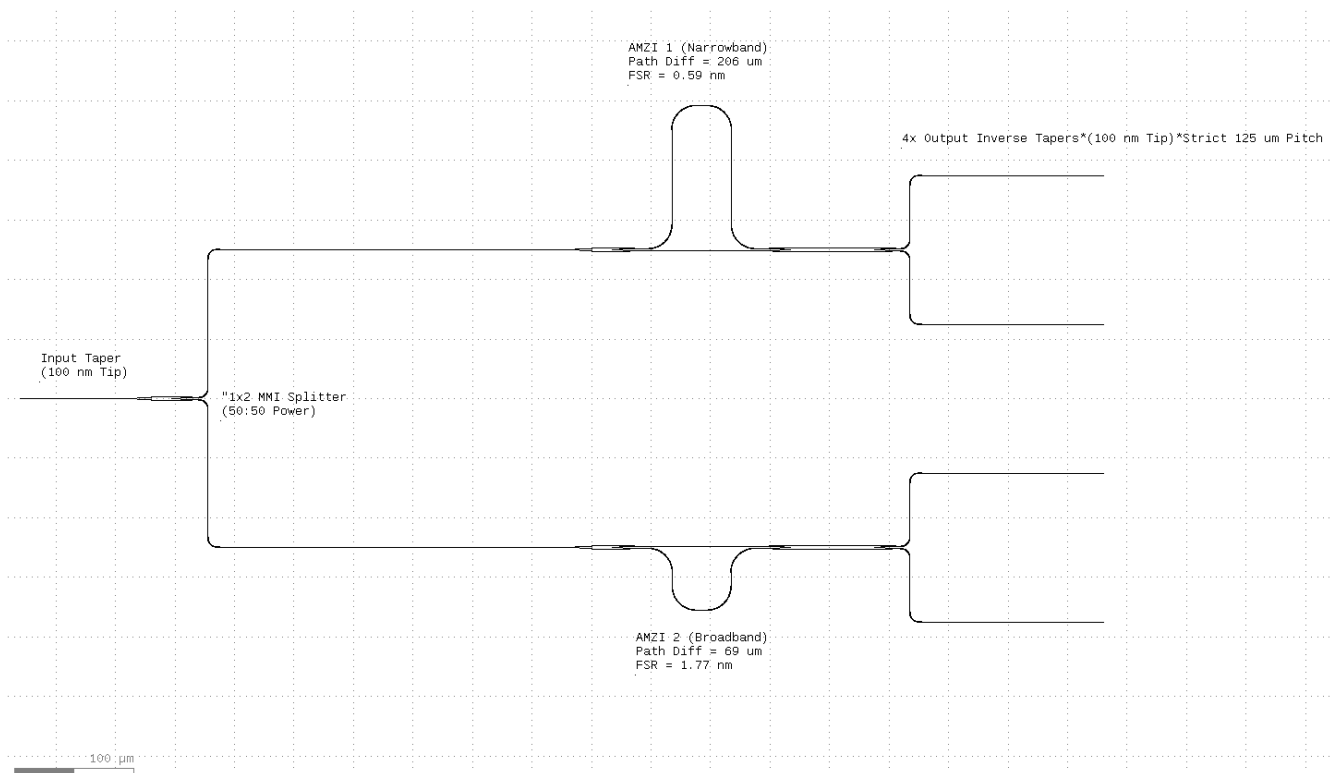


Figure 4.14: GDSII mask layout of the dual-AMZI PIC, demonstrating symmetric routing at 679 nm.

For a strict single TE mode operation, structures (components) were designed using KLayout in conjunction with the Python-based gdsfactory framework as shown in Fig. 4.14. Leveraging the integrated CORNERSTONE Process Design Kit (PDK) for the 200 nm SiN visible platform, the core waveguide geometry was constrained to an exact width of 380 nm for waveguides and 3.5 μm for 1x2 and 2x2 MMI box body. For edge coupler, total length was set at 400 μm and narrow tip width at 100 nm.

After designing the circuit to make sure the circuit should be connected and tested as per standards in lab . to get this circuit output four inverse tappers spaced 125 μm apart. this distance provides the good room to connect fibers and connectors directly to the circuit ex: connectors like fiber array units (FAUs) and standard multi-mode fibers .

4.4 Results - EW

Lumerical scripts were developed. Scripts processed simulation data to generate graphs characterising behaviour and to output data in tabular form. Data was transferred to an Excel spreadsheet. Each row recorded ($\lambda, P_1, P_2, P_3, P_4$) where $P_1 - P_4$ were simulation optical output powers from the AMZIs. Excel formulae calculated error signal and error signal slope for each wavelength. In combination with photodiode datasheet data, Excel formulae also calculated 1σ noise values and discrimination sensitivity row-by-row for each wavelength λ . This format was later useful for collision analysis (see 4.2.2 and 4.4.8).

Table 4.3 gives a summary of the circuit-level results derived from Excel and downstream VBA processing. Noise modelling affected all metrics, so the noise model is discussed first below, then a section for each of the other headline metrics follows.

Metric name	10Hz value	100Hz value	1000Hz value
Wavelength uncertainty range (fm)	0.14 - 0.67	0.45 - 2.11	1.43 - 6.66
3σ	0.42 - 2.01	1.35 - 6.33	4.29 - 19.98
E_1 error signal uncertainty range (x 10⁻⁶)	0.26 - 0.83	0.09 - 2.64	0.28 - 8.35
3σ (x 10⁻⁶)	0.78 - 2.49	0.27 - 7.92	0.84 - 25.05
E_2 error signal uncertainty range (x 10⁻⁶)	0.03 - 0.82	0.1 - 2.6	0.33 - 8.21
3σ (x 10⁻⁶)	0.09 - 2.46	0.3 - 7.8	0.99 - 24.63
Minimum detectable wavelength change (fm)	2	9.4	30
Maximum detectable wavelength range	No collisions 679nm $\pm$ 0.5nm	No collisions 679nm $\pm$ 0.5nm	No collisions 679nm $\pm$ 0.5nm
Discrimination sensitivity (minimum, pm⁻¹)	1502	475	150

Table 4.3: PIC results summary

4.4.1 Noise analysis approach - EW

The analysis was based on Hayden et. al.³⁸. 1σ values for wavelength uncertainty and error signal uncertainty were derived. 3σ was chosen as an operational target when stating device-level wavelength discrimination capability. $\pm 3\sigma$ limits are a standard convention in statistical process control, as stated in Section 4.1.

4.4.2 Assumptions - EW

The analysis is restricted to photodiode noise only. Some noise sources have been excluded, see Table 4.4.

Assumption #	Assumption	Rationale
1	Residual laser current noise.	Most is common-mode cancelled in normalised error signals.
2	Downstream electronics noise.	Depends entirely on component choice.
3	Internal PIC noise.	It would be difficult to justify any model as this device won't be fabricated

Table 4.4: Noise model assumptions

The photodiode stage was not excluded because photodiodes could be integrated with the PIC for an off-the-shelf component - its datasheet would stop at the O2E conversion stage. Including the photodiode stage yields a complete picture of the opto- part of an optoelectronic circuit. A candidate photodiode was selected for analysis purposes (see Appendix A). Noise contributions were modelled from its datasheet.

Electronic detection bandwidth was treated parametrically. The analysis calculated total RMS noise for bandwidths of 10, 100 and 1000Hz. Gaussian noise was assumed with noise power scaling $\approx \sqrt{\text{Bandwidth}}$. Any figures used in discussion in this report are based on a bandwidth of 100Hz. Table 4.3 gives a summary for all three bandwidths.

4.4.3 RMS output power uncertainty - EW

Post-photodiode, the uncertainty in power P_i at output i has contributions from signal shot noise and dark current shot noise. See equation 4.3

$$\sigma_{P_i} = \sqrt{\sigma_{shot_i}^2 + \sigma_{dark_i}^2} \quad (4.3)$$

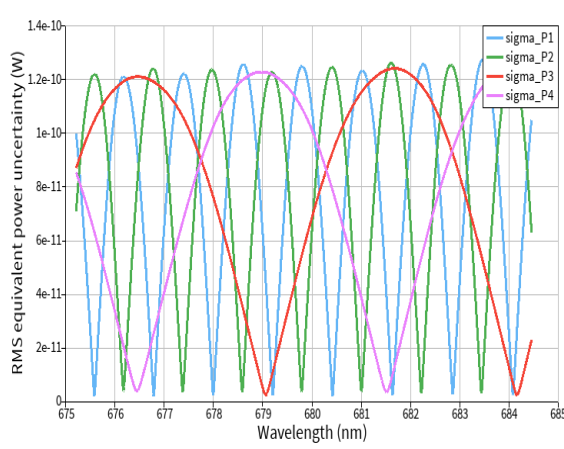
Shot noise RMS current is given by $\sqrt{2 \cdot q \cdot I \cdot B}$, where q is electron charge, I is signal or dark current and B is the bandwidth. Dividing by responsivity R gives RMS noise power.

$$\sigma_{P_{dark}} = \sqrt{\frac{2qP_{dark}B}{R}} \quad \sigma_{P_{signal}} = \sqrt{\frac{2qP_{signal}B}{R}} \quad (4.4)$$

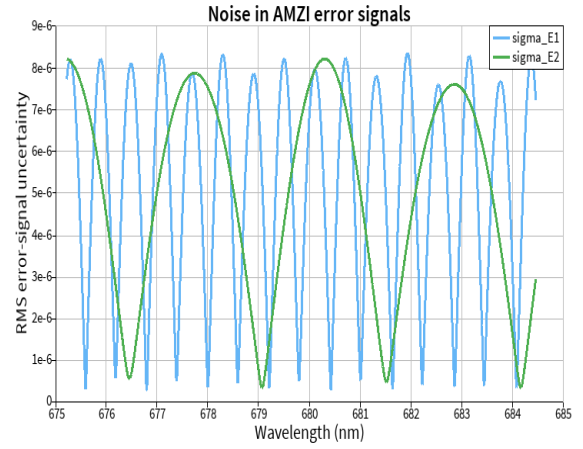
Post-photodiode, the uncertainty in power for each circuit output was calculated in Excel for all wavelength points. It's shown in Fig. 4.15a. Output power was sinusoidal: processing this through RMS produced a series of positive lobes.

4.4.4 Error signal uncertainty

To model noise-induced uncertainty in an error signal, recall that noise presents as a small fluctuation in output power.



(a) RMS uncertainty at PIC outputs



(b) Uncertainty propagated to error signals

Figure 4.15: Noise impacts photodiode outputs and processed error signals

$$E = \frac{P_1 - P_2}{P_1 + P_2} \quad \therefore \quad dE \approx \frac{\partial E}{\partial P_1} dP_1 + \frac{\partial E}{\partial P_2} dP_2 \quad (4.5)$$

In equation 4.4.4, dE, dP_1, dP_2 are the noise-induced fluctuations in error signal and the power values E is calculated from. The equation is a first-order Taylor approximation. Partial derivatives are obtained by treating one output power as constant and differentiating with respect to the other, eg $\frac{\partial E}{\partial P_1} = \frac{2P_2}{(P_1 + P_2)^2}$. Noise is assumed random so the contribution at each AMZI output is independent. Holding $dP_2 = 0$ gives

$$dE \approx \frac{\partial E}{\partial P_1} dP_1$$

If σ_{dP_1} is the standard deviation in dP_1 then the standard deviation in dE fluctuations due to noise at output P_1 is:

$$\sigma_{dE_{P_1}} = \left| \frac{\partial E}{\partial P_1} dP_1 \right| \sigma_{dP_1} \quad (4.6)$$

We could write a similar expression for $\sigma_{dE_{P_2}}$. Squaring gives the variance; dP_1, dP_2 are independent so their variances add. The standard deviation (RMS uncertainty) in error signal E_1 due to noise in P_1, P_2 is:

$$\sigma_{dE_{P_1}} = \sqrt{\left(\frac{\partial E}{\partial P_1} \right)^2 \sigma_{P_1}^2 + \left(\frac{\partial E}{\partial P_2} \right)^2 \sigma_{P_2}^2} \quad (4.7)$$

And again a similar expression can be written for RMS uncertainty in the error signal E_2 (derived from the second AMZI). All of these values were calculated for each wavelength by formulae in the Excel analysis sheet.

Noise in error signals is shown in Fig. 4.15b. The noise contributions form a series of positive lobes because the derivative of cosine is sine, and values are squared in processing.

4.4.5 Minimum detectable wavelength change - EW

Two metrics are given. The first is an estimate of uncertainty in wavelength measurement if both E_1, E_2 are used together. It is used later to demonstrate a sensitivity for the device as a whole. The second gives the smallest wavelength drift that at least one of the AMZIs can reliably detect.

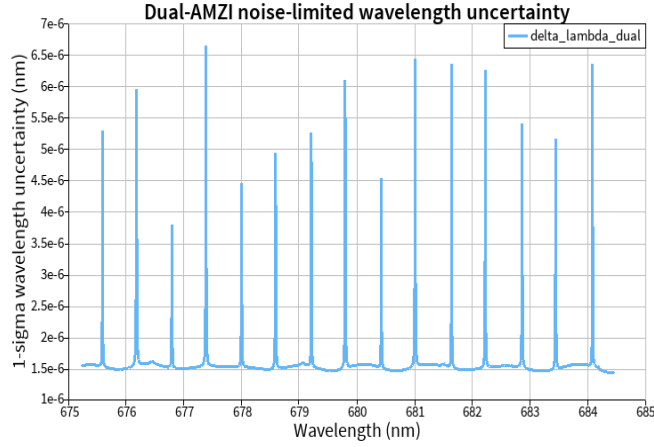


Figure 4.16: 1 σ wavelength uncertainty

Device-level 1 σ wavelength uncertainty

For a small wavelength change

$$\Delta E \approx \frac{dE}{d\lambda} \Delta \lambda \quad (4.8)$$

So the wavelength change corresponding to 1 σ movement in error signal for one AMZI is

$$\sigma_\lambda \approx \frac{\sigma_E}{|dE/d\lambda|}$$

This can't simply be combined with the equivalent expression for the other AMZI, because they are two measurements derived from the same wavelength. Taking the reciprocal of equation instead measures how strongly an AMZI responds, compared to its noise, for a unit wavelength change. It measures how much useful wavelength information each AMZI contributes. We *can* combine these values and obtain the 1 σ uncertainty in the wavelength estimate when both error signals are used together.

$$\frac{1}{\sigma_{\lambda_{dual}}^2} = \frac{1}{\sigma_{\lambda_{AMZI1}}^2} + \frac{1}{\sigma_{\lambda_{AMZI2}}^2} \quad (4.9)$$

Clearly the noise impact is wavelength-dependent, even though a single noise figure was used from the photodiode datasheet. This is illustrated in Fig. 4.16. For most wavelengths, uncertainty is around 1 $\sigma \approx 1.5 fm$; spikes up to 1 $\sigma \approx 6.6 fm$ occur.

AMZI-level 3 σ wavelength uncertainty

When are two wavelengths λ_a, λ_b indistinguishable given both (E_1, E_2) pairs have uncertainty? Each error signal is noisy, so when comparing the two E_1 signals, their difference is:

$$\sigma_{\Delta E_1} = \sqrt{\sigma_{E_1\lambda_a}^2 + \sigma_{E_1\lambda_b}^2}$$

Similarly for E_2 . This comes from adding the variances of two independent noisy measurements. This analysis is for very nearby wavelengths: assume that the wavelength noise values are very similar to the E_1 noise value, or $\sigma_{\Delta E_1} \approx \sigma_{E_1\lambda_a} \approx \sigma_{E_1\lambda_b}$. Then

$$\sigma_{\Delta E_1} \approx \sqrt{2} \sigma_{E_1}$$

Similarly again for E_2 . Combining, the distance between two error signal points must be at least 3σ to be reliably distinguishable:

$$\Delta E_{min} = 3\sqrt{2}\sigma_E$$

Referring back to equation 4.8, the wavelength change generating a 3σ response in the error signal output is

$$\Delta\lambda_{min} = \frac{3\sqrt{2}\sigma_E}{dE/d\lambda} \quad (4.10)$$

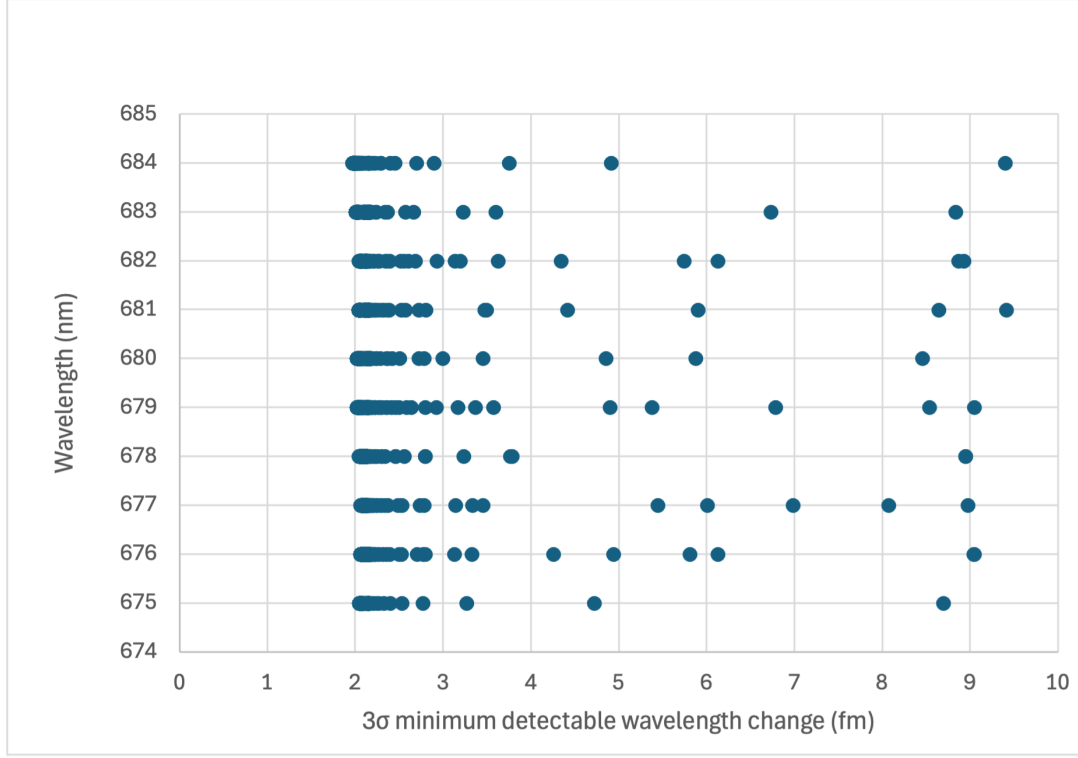


Figure 4.17: Wavelength vs 3σ error uncertainty 674-684nm, 100Hz

Note that this is the minimum detectable wavelength change.

Figure 4.17 shows the value for the most sensitive AMZI at each wavelength over a broad wavelength range, operating at 100Hz electronic detection bandwidth. Each point asks: at this wavelength, what is the smallest wavelength change that at least one AMZI can distinguish at 3σ ?

Figure 4.18 gives a more detailed view over a shorter wavelength range 678 – 681nm and for a bandwidth of 10Hz. For most wavelengths, the best performing AMZI resolves changes of around 0.6–0.7fm. At 679 nm, that corresponds to roughly 0.4–0.45 MHz. Periodic spikes rise to about 3fm - roughly 2 MHz. These occur periodically because AMZI1 approaches a sensitivity minima and the device relies on AMZI2 with its shallower error signal slope.

Assuming an electronic detection bandwidth of 10Hz, the device is sensitive to wavelength changes of the order 0.4 – 2 MHz. This is very close to the target set by Requirement #2 (see 4.1.1 section). However, 10Hz may not be suitable for laser stabilisation systems.

4.4.6 Discrimination sensitivity - EW

A metric was developed to characterise noise impact on the operational output - an error signal pair. Consider (E_1, E_2) as a point. After scaling each by its own noise standard deviation, the wavelength-induced displacement is

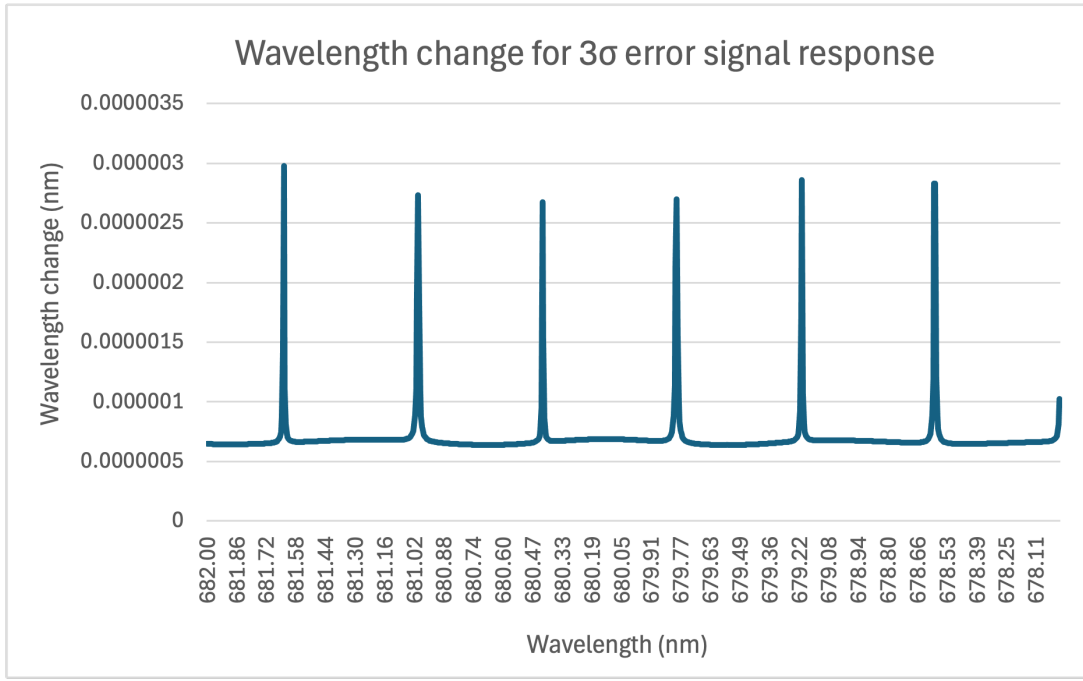


Figure 4.18: Wavelength vs 3σ error uncertainty 678-681nm, 10Hz

$$\left(\frac{\Delta E_1}{\sigma_{E_1}}, \frac{\Delta E_2}{\sigma_{E_2}} \right)$$

Recall ??; $dE \approx \frac{dE}{d\lambda} d\lambda$. The axes (error signals) are independent, so the displacement ds due to noise is:

$$ds = \sqrt{\left(\frac{dE_1/d\lambda}{\sigma_{E_1}} d\lambda \right)^2 + \left(\frac{dE_2/d\lambda}{\sigma_{E_2}} d\lambda \right)^2} \quad (4.11)$$

Factoring out $d\lambda$, noise-normalised displacement (*discrimination sensitivity* S_λ) per unit wavelength is:

$$S_\lambda = \frac{ds}{d\lambda} = \sqrt{\left(\frac{dE_1/d\lambda}{\sigma_{E_1}} \right)^2 + \left(\frac{dE_2/d\lambda}{\sigma_{E_2}} \right)^2} \quad (4.12)$$

This was calculated for each wavelength point in the evaluation band and expressed in units of pm^{-1} . A worst-case for each electronic detection bandwidth is reported in table 4.3.

4.4.7 Signal to noise ratio - EW

Following Hayden³⁸, the analysis has obtained noise power, detection limits and a sensitivity measure. Hayden's work defines a limit of detection rather than a signal to noise ratio, but this analysis does offer SNR. For each AMZI,

$$SNR_i = \frac{|\Delta E_i|}{\sigma_{E_i}}$$

Recall again 4.8:

$$SNR_i = \frac{|dE/d\lambda| \Delta\lambda}{\sigma_{E_i}}$$

This term appears in 4.11, so

$$SNR_{dual} = \sqrt{SNR_1^2 + SNR_2^2}$$

and

$$SNR(\Delta\lambda) = S_\lambda \Delta\lambda \quad (4.13)$$

Being based on the linear approximation 4.8, this is valid for small wavelength deviations where the error signal curve is approximately straight.

4.4.8 Unambiguous range around target - EW

With $FSR_2 = 5.3$ (see section 4.2.1), INTERCONNECT simulation generated 2001 data points between 674-684nm. Error signal slope, 1σ noise and 3σ limits were calculated. The VBA macro developed to investigate candidate phase offset differences (see section 4.2.2) was repurposed. The macro mined the spreadsheet for wavelength collisions between 678 and 680nm. None were found. This could be because the analytical selection of ϕ_2 was in fact correct (see section 4.2.1). But it could also be because none of the 2001 points chosen did collide exactly.

Another analysis ran using equations 4.1 to generate AMZI outputs, that is, not using the wavelength sample points selected in INTERCONNECT. A single pair of colliding wavelengths was found, 678.396 and 679.606nm. This yielded a collision-free tolerance of $679nm \pm 0.60635nm$ - see Fig. 4.19. Requirement #1 in Section 4.1.1 is met.

Note that this analysis was also performed for $FSR_2 = 5.1nm$. The results were not as good as $FSR_2 = 5.3nm$. The intention had been to check other FSR_2 values but time did not permit - again, please see section 4.2.1.

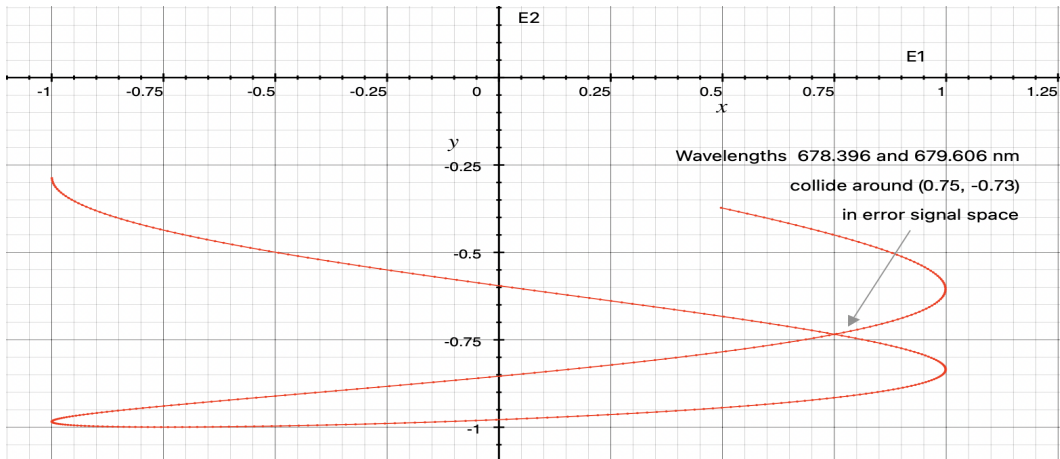


Figure 4.19: Collisions around 679 nm

4.4.9 Core Figures of Merit - PS

A Figure of Merit(FOM) gave a single and overall estimate check and optimize 2x2 MMI. it evaluates the device based on its transmission ,balance, and to secure light leakage. for 2x2 MMI it consist port 1 and port2 as inputs and port 3 and port 4 as output. Figure of merit is calculated using the following five checks.³⁷.

- **Total Light loss(Insertion Loss):**It checks how much light reaches the output ports. When light enters to input 1 ,to measure the output formula for that $-10\log_{10}(|S_{31}|^2 + |S_{41}|^2)$. the perfect structure has 0 db loss means no light is loss inside the device.

- **Even Splitting (Power Imbalance):**it ensures that light splits evenly between two ports and it calculated using formula $10\log_{10}(|S_{31}|^2/|S_{41}|^2)$.which means a perfect 50/50 splitters gives 0 db loss.
- **Phase Imbalance (Phase Error):**By structure ,this device made to push the light waves 90° out of sync when they exit which means naturally it creates 90° ($\pi/2$) phase difference with its output, which also helps to track unwanted disturbance from that 90° shift which calculated using formula $|\angle S_{31} - \angle S_{41}| - 90^\circ$.
- **Reflection (Return Loss):**It helps to detect the amount of light bounce back into the input port and formula for that is $-10\log_{10}(|S_{11}|^2)$.
- **Crosstalk (Isolation):**This helps to measure the how much amount of light accidentally leaks from sideways from the wrong unused port. Formula to calculate $-10\log_{10}(|S_{21}|^2)$.

To achieve the best possible PIC deign, we collect and combine all metrics in single final FOM equation. this equations makes sure that it allows the most light through ,and removes point for physical errors or leaks as follows:

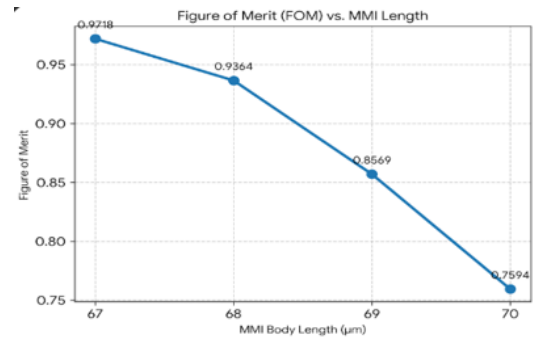
$$FOM = \frac{|S_{31}|^2 + |S_{41}|^2}{1 - \alpha ||S_{31}|^2 - |S_{41}|^2| - \beta |\Delta\phi - \frac{\pi}{2}|} \quad (4.14)$$

In this equation, α and β are weighting coefficients. If the downstream photonic circuit is highly sensitive to power mismatch, we increase α . If the MMI is acting as a beam splitter for an interferometer where exact phase is critical, we increase β to force the FDTD solver to prioritize a perfect 90° phase offset over raw transmission efficiency.

To extract a common FoM factor, we put all the data for its parameters in Fig. 4.20a and a graph depicting transmission over length sweeps of 2x2 MMI body in Fig. 4.20b.

MMI body Length(um)	Total Transmission	Inse rtion Loss	Total transmitted power	Excess Loss	Splitting power imbalance	Phase differe nce
70 with source	40.3% at each output	0.9d B	80.6%	-	-	90
70	Port3=45.2% Port4=41.8%	0.60 dB	87%	0.604dB	0.34dB	92.44°
69	Port3=45.5% Port4=47.7%	0.31 dB	93.2%	0.306dB	0.205dB	91.78°
68	Port3=48.0% Port4=48.7%	0.15 dB	96.7%	0.146dB	0.063dB	90.95°
67	Port3=49.1% Port4=49.7%	0.05 dB	98.8%	0.054dB	0.052dB	89.76°

(a) Core Figures of Merit parameters for 2x2 MMI over different lengths



(b) Figure of Merit (FOM) vs. MMI Length-2x2 MMI

Figure 4.20: Core Figures of Merit for the 2x2 MMI

Fundamental Component performance - AL

To make sure that the final circuit is successful we tested and fine-tuned each individual part at 679 nm wavelength using lumerical software³⁹ .also measures the splitters by calculating the insertion loss by adding together the power coming out of both exit arms.this single calculation already includes every type of power loss or all the light lost inside the device this includes the natural splitting loss and any extra scattering,as it is already includes in insertion loss number³⁷ .

4.4.10 Impact of Manufacturing Variations

the exact size and shape of the waveguide control how well it operates the design in other words waveguides are critical for guiding light efficiently⁴⁰ .as per manufacturing guidelines from cornerstone the thickness of waveguide accuracy $\pm 15\text{nm}$ ⁴¹ . while cornerstone do not provide an exact error limit for width, we assume standard account

Component	Total Transmission	Insertion Loss	Power Imbalance	Phase Difference
1x2 MMI (16.5 μm body)	P2=49.7%, P3=49.7% (Total 99.5%)	0.023 dB	-0.0001 dB	0.011°
1x2 MMI (17.5 μm body)	P2=48.7%, P3=48.7% (Total 97.4%)	0.115 dB	-0.0012 dB	0.008°
Inverse Taper	92.2%	0.354 dB	-	-
10 μm Straight Waveguide	100.0%	~ 0 dB	-	-
S-Bend (50 μm length)	98.9%	0.046 dB	-	-
90° Bend (50 μm radius)	99.9%	0.005 dB	-	-

Table 4.5: Figures of Merit for 1x2MMI, Inverse Taper, Waveguide, S-bend & 90-bend waveguide at 679 nm

for manufacturing drift which is ± 20 nm. to ensure that our $380\text{nm} \times 200\text{nm}$ design works well, we ran simulations testing this setup across the max physical limits of dimensions (testing thickness from 185 to 215 nm and width from 360 to 400nm).

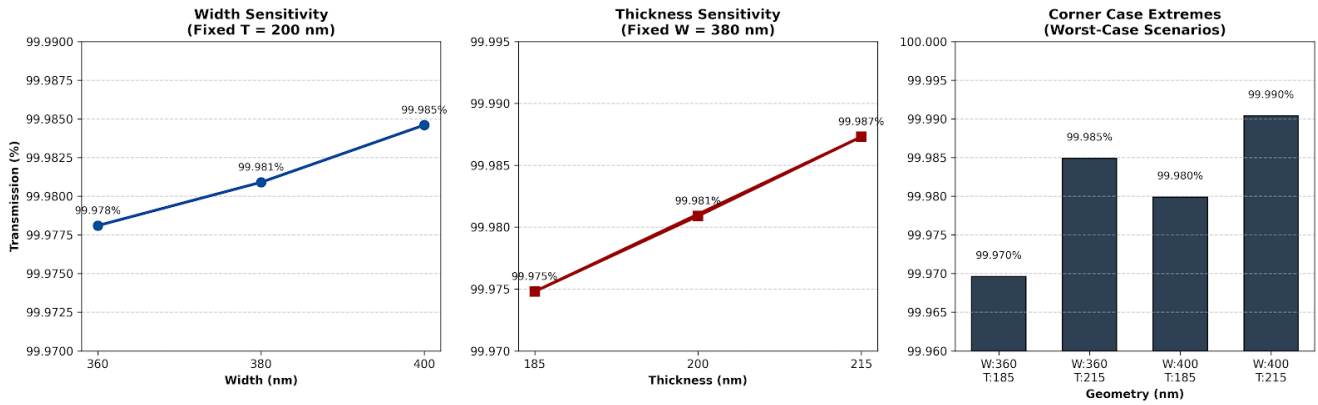


Figure 4.21: Manufacturing Variations analysis showing highly stable waveguide transmission across width variations (left), thickness variations (center), and extreme corner cases (right).

The waveguide design is highly stable. Whether we change the thickness or width individually, transmission of light always remained above 99.97%, even when we combined the worst case scenarios for width and thickness limits (360 and 185 nm respectively).

Going ahead, the light stayed trapped in the core. All the tested simulations for the waveguide give the same transmission, and the light profile looks the same for every waveguide variation. Therefore, there is no risk of the light leaking out. This ensures that our target wavelength will perform perfectly through the circuit regardless of small manufacturing differences⁴⁰.

4.5 Discussion - EW

4.5.1 Are requirements met?

Please see the Requirements section 4.1.1.

Requirement #1 In section 4.4.8, an unambiguous wavelength range centred exceeding $679\text{nm} \pm 0.5\text{nm}$ was demonstrated. This requirement is met.

Requirement #2 In section 4.4.5, wavelength resolution of the order of $0.4 - 2$ MHz was demonstrated at 10Hz electronic detection bandwidth. This requirement is close to being met, but the bandwidth might be unsuitable for a laser stabilisation application. On balance, performance justifies future work to further improve the device for the target application.

Requirement #3 The design used a normalised error signal to give not just the magnitude of error (laser wavelength drift), but also its "direction" (whether it had increased or decreased). This requirement is met.

4.5.2 Design choice - EW

A normalised error signal permits rapid downstream laser current control: the sign of the error signal already indicates which way the current needs to be adjusted. This requires two error values. This added design complexity. However, using two AMZIs improved the unambiguous wavelength range (at least in principle). The additional design complexity was justified.

A Fabry-Perot cavity can replace one of the AMZIs. However, this is more complex to design and fabricate. This option was explored early in the design cycle, but discounted as unnecessarily complex.

Another control method is "hill climbing" control. This uses a single error value: the magnitude of the value indicates how far the laser has drifted, but there is no information about whether it has drifted up or down in wavelength. The design is simpler, but the control method is more complex. It's necessary to "guess" whether a positive or negative adjustment is needed. By continually tracking the results of adjustments, it's obvious whether the guess was wrong and the opposite action can be taken. A signed error signal was the preferred choice.

4.5.3 Suitability for precision atomic physics applications

The PIC could be employed for wavelength-sensing in a compact laser-locking system - especially if the wavelength discrimination can be further improved. Applications could include strontium-based atomic clocks, cold-atom experiments and other precision photonic systems. The PIC wouldn't replace a strontium vapour cell - but it might be developed to monitor and stabilise the laser that drives the strontium transition.

The design aimed to specify a device that could detect wavelength-equivalent frequency changes of around the $^3P_0 \rightarrow ^3S_0$ transition linewidth. With development, if noise could be controlled adequately - and if the surrounding electronic circuit were fast enough - then this PIC could reliably resolve wavelength changes of a similar order.

4.6 Future Work - EW

Key results - EW

- Collision-free tolerance of $679\text{nm} \pm 0.60635\text{nm}$ (a 1.21nm unambiguous wavelength span) around 679nm.
- Wavelength resolution potential to 0.4MHz demonstrated at 10Hz electronic detection bandwidth.

Developing the PIC further: follow-on work ideas - EW

- The error signal slope at $FSR_2 = 5.3$ was well within noise tolerances; FSR_2 could probably be increased further. This should be investigated as it might increase the collision-free range of around the target wavelength. It's easy to repeat the simulation and extract results thanks to the Lumerical scripts developed. Results can be (and were) imported into Excel easily, where formulae derive error signal slope and standard deviations. The VBA macro to determine collision free range around target is in place. A thorough (FSR_2) would have been undertaken in this way and using this processing architecture, but the team ran out of time.
- The ϕ_2 setting could be optimised further with finer resolution of wavelength points used as input the the VBA analysis macro.
- Wavelength discrimination sensitivity is good, but could perhaps be improved further. This work would start by decreasing FSR_2 , as its sensitivity seemed to be the limiting factor (see section 4.4.5).
- It would be useful to extend the noise analysis to include realistic values for downstream electronics. This would inform another iteration of PIC design, to boost PIC performance enough to give excellent sensitivity and resolution in a real-world operational content.

- The PIC circuit could be enhanced with an integrated temperature sensor. During manufacture the PIC could be precalibrated at different ambient temperatures to install the mapping $(\lambda, T) \rightarrow (E_1, E_2)$. The device then would ship ready for use and offering robust performance under environmental temperature changes.

5 Conclusion

This work investigated two complementary approaches to the miniaturisation of optical systems: the development of a compact free-space experimental platform and the design and simulation of a photonic integrated circuit. The overall objective was to assess whether useful optical functionality could be preserved while reducing physical footprint, cost, and dependence on conventional laboratory-scale equipment.

The free-space platform successfully demonstrated both quantitative laser characterisation and fundamental optical experimentation. Knife-edge profiling produced an error-function response with $R^2 \approx 0.995$, supporting the approximation of the transverse laser profile as Gaussian. The reconstructed distribution gave a $1/e^2$ beam radius of approximately 0.281 mm, corresponding to a beam diameter of 0.563 mm and an intensity FWHM of approximately 0.331 mm. Interference was also successfully obtained using both Michelson and Mach–Zehnder configurations. The presence of distinguishable fringes indicated sufficient spatial overlap, coherence and optical-path stability for interferometric measurements, while the MZI further demonstrated the expected dependence of interference on the relative polarization of the two optical paths. These results show that the platform could reproduce standard optical characterisation and interferometric experiments despite its reduced dimensions and the use of low-cost custom mounting hardware.

An important consideration in miniaturising a free-space optical system is whether its reduced dimensions adversely affect stability and repeatability. The compact arrangement places optical components in close proximity and reduces the available range for alignment, while vibrations or small mechanical displacements may produce significant changes in an interference pattern. Consequently, the quality and persistence of the observed interference fringes, the ability to reconstruct different experimental arrangements on the same platform, and the response of the system to mechanical disturbances were considered when assessing the suitability of the miniature platform. These measurements were supplemented by consideration of the cost and physical footprint of the developed system relative to conventional laboratory equipment.

The results nevertheless highlighted the trade-offs introduced through miniaturisation. Shorter component separations reduced the available space for alignment, making the system comparatively sensitive to small angular and positional errors. Manufacturing tolerances in the custom mounts similarly limited the precision obtainable relative to commercial optical hardware. However, the persistence of interference and the ability to reconstruct different experimental configurations indicate that these limitations did not prevent the platform from performing its intended functions. The system therefore represents a practical compromise between optical performance, accessibility, portability and cost.

The PIC study extended the investigation towards a substantially greater degree of miniaturisation. Individual silicon nitride photonic components were designed and characterised before being incorporated into a dual-AMZI architecture for wavelength discrimination around 679 nm. The differing free spectral ranges of the two interferometers provided complementary error signals, reducing the wavelength ambiguity associated with a single periodic interferometer response.

Collision analysis identified an unambiguous operating tolerance of approximately $679 \text{ nm} \pm 0.60635 \text{ nm}$, corresponding to a collision-free wavelength span of approximately 1.21 nm. Behaviour was modelled in the presence of realistic photodiode noise and the architecture was capable of discriminating wavelength-equivalent frequencies of around 0.4 – 2 MHz. This is of the order of the $^3P_0 - ^3S_0$ repumping Sr transition.

These are encouraging results: they establish a simulated proof of concept for the proposed wavelength-discrimination architecture. Such an architecture might be suitable for use in atomic physics applications or precision photonic systems with further development and refinement. The results also demonstrate that interferometric functionality can be transferred from bulk free-space optics to an integrated photonic platform.

Combined together, the two parts demonstrate a progression in optical miniaturisation. The free-space system retained conventional optical components while reducing the scale and cost of their supporting infrastructure, whereas the PIC transferred related interferometric principles directly onto an integrated platform. The experimental and simulated results therefore support the central objective of this investigation: that useful optical functionality

can be retained across progressively smaller implementations, from a portable experimental system for optical characterisation and interferometry to an integrated photonic architecture intended for wavelength discrimination and stabilisation.

A Appendix

A.1 Electronic components

The following were selected as representative components for a final optoelectronic circuit containing the PIC.

- Fibre-pigtailed laser: ThorLabs LPS-675-FC.
- Photodiode: ThorLabs FDS100.
- Photodiode amplifier: DLPCA-200 Variable Gain Low Noise Current Amplifier FEMTO Messtechnik GmbH Klosterstr. 64 10179 Berlin Germany.
- Data acquisition: Diligent MCC USB-231 USB DAQ.

A.2 Voltage Vs Polarization Plot

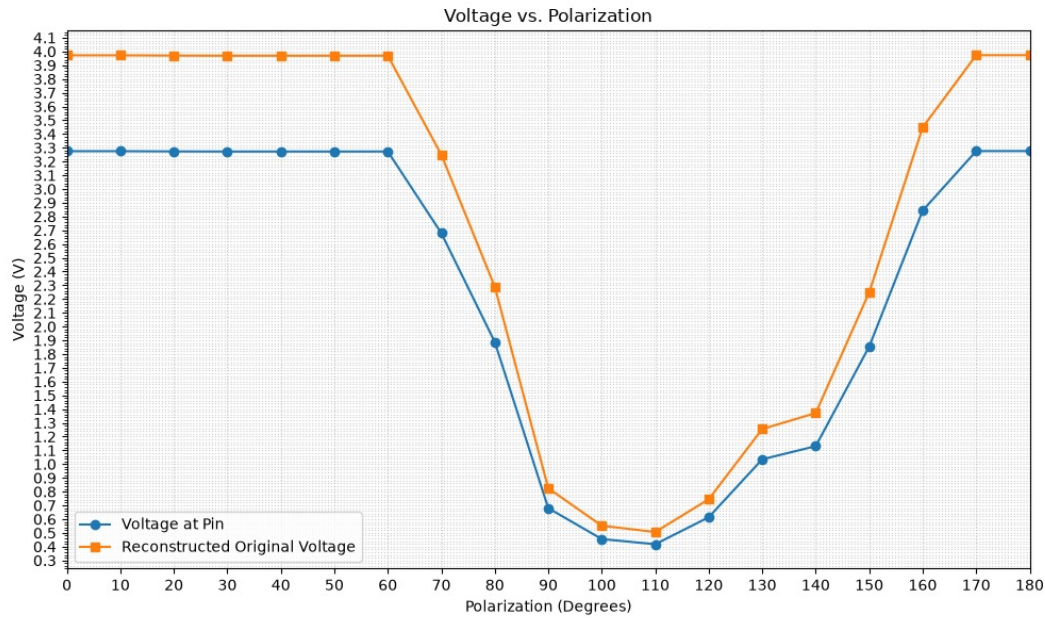


Figure A.1: Photodiode voltage measured over 10° polarization increments.

A.3 Summary of resistor changes for MAX1978 board

Table A.1: Default and proposed MAX1978 EV-kit resistor configuration.

Component	Function	Default configuration	Proposed configuration	Resulting limit
R_2	MAXV upper	49.9 k Ω	Swap, 60.4 k Ω	$ V_{\text{TEC}(\text{MAX})} \approx 3.75 \text{ V}$
R_3	MAXV lower	100 k Ω	No change	Forms MAXV divider
R_4	MAXIN upper	Short connection	Open connection, add 16.2 k Ω	$I_{\text{TECN}(\text{MAX})} \approx -1.9 \text{ A}$
R_5	MAXIN lower	Not present/open	Add 100 k Ω	$I_{\text{TECN}(\text{MAX})} \approx -1.9 \text{ A}$
R_6	MAXIP upper	Short connection	Open connection, add 16.2 k Ω	$I_{\text{TECP}(\text{MAX})} \approx +1.9 \text{ A}$
R_7	MAXIP lower	Not present/open	Add 100 k Ω	$I_{\text{TECP}(\text{MAX})} \approx +1.9 \text{ A}$
R_1	Current-sense	0.068 Ω	No change	Used in current limiting

References

- [1] Oskay WH, Diddams SA, Donley EA, Fortier TM, Heavner TP, Hollberg L, et al. Single-Atom Optical Clock with High Accuracy. *Phys Rev Lett*. 2006 Jul;97:020801. Available from: <https://link.aps.org/doi/10.1103/PhysRevLett.97.020801>.
- [2] Yan Y, Zheng Y, Sun H, Duan J. Review of Issues and Solutions in High-Power Semiconductor Laser Packaging Technology. *Frontiers in Physics*. 2021 May;9:669591. Available from: <https://doi.org/10.3389/fphy.2021.669591>.
- [3] Zhao YN, Zhang J, Stuhler J, Schuricht G, Lison F, Lu ZH, et al. Sub-Hertz frequency stabilization of a commercial diode laser. *Optics Communications*. 2010 Dec;283(23):4696-700. Available from: <https://doi.org/10.1016/j.optcom.2010.06.079>.
- [4] Dimarcq N, Gertsvolf M, Mileti G, Bize S, Oates CW, Peik E, et al. Roadmap towards the redefinition of the second. *Metrologia*. 2024 Feb;61(1):012001.
- [5] Zuo Y, Dai S, Cao S, Chen W, Liu K, Zheng F, et al. Towards a High-Performance Optical Clock Based on Single 171-Yb Ion. In: 2021 IEEE 6th Optoelectronics Global Conference (OGC); 2021. p. 91-3.
- [6] Hecht E. *Optics*. 5th ed. New York, NY, USA: Pearson; 2017.
- [7] Paschotta R. Coherence. RP Photonics AG;. Available online at <https://www.rp-photonics.com/coherence.html>. RP Photonics Encyclopedia. Available from: <https://www.rp-photonics.com/coherence.html>.
- [8] Konijnenberg S, Adam AJL, Urbach HP. *BSc Optics*. 2nd ed. Delft, Netherlands: TU Delft OPEN Publishing; 2024.
- [9] Paschotta R. Polarization of Light. RP Photonics AG; 2024. Available online at https://www.rp-photonics.com/polarization_of_light.html. RP Photonics Encyclopedia. Available from: https://www.rp-photonics.com/polarization_of_light.html.
- [10] Saleh BEA, Teich MC. *Fundamentals of Photonics*. 3rd ed. Wiley; 2019.
- [11] Paschotta R. Laser Beam Characterization. RP Photonics AG;. Available online at https://www.rp-photonics.com/laser_beam_characterization.html. RP Photonics Encyclopedia. Available from: https://www.rp-photonics.com/laser_beam_characterization.html.
- [12] Paschotta R. Interferometers. RP Photonics AG;. Available online at <https://www.rp-photonics.com/interferometers.html>. RP Photonics Encyclopedia. Available from: <https://www.rp-photonics.com/interferometers.html>.
- [13] Paschotta R. Optical Clocks. RP Photonics AG;. Available online at https://www.rp-photonics.com/optical_clocks.html. RP Photonics Encyclopedia. Available from: https://www.rp-photonics.com/optical_clocks.html.
- [14] Ludlow AD, Boyd MM, Ye J, Peik E, Schmidt PO. Optical atomic clocks. *Reviews of Modern Physics*. 2015;87:637-701.
- [15] Gillen GD. Laser Laboratory Beam Alignment Skills: Course Package. arXiv preprint arXiv:240416225. 2024. Available from: <https://arxiv.org/abs/2404.16225>.

- [16] EPCOS AG. NTC Thermistors for Temperature Measurement: Miniature Sensors with Bendable Wires, Series B57861S; 2018. Datasheet. Available from: <https://docs.rs-online.com/74a8/0900766b813c0e6b.pdf>.
- [17] Steinhart JS, Hart SR. Calibration Curves for Thermistors. Deep-Sea Research and Oceanographic Abstracts. 1968 Aug;15(4):497-503.
- [18] Directly Heated Negative Temperature Coefficient Thermistors – Part 1: Generic Specification. Geneva, Switzerland; 2008.
- [19] Renesas Electronics Corporation. RA4M1 Group Datasheet; 2023. R01DS0355EJ0110, Rev. 1.10. Available from: <https://www.renesas.com/en/document/dst/ra4m1-group-datasheet>.
- [20] Maxim Integrated Products, Inc . MAX1978/MAX1979 Integrated Temperature Controllers for Peltier Modules; 2010. Datasheet. Available from: <https://www.analog.com/media/en/technical-documentation/data-sheets/max1978-max1979.pdf>.
- [21] Wang W, Subhankar S, Britton JW. A practical guide to feedback control for Pound-Drever-Hall laser linewidth narrowing; 2025. Available from: <https://arxiv.org/abs/2412.04635>.
- [22] Kozawa Y, Sato S. Single higher-order transverse mode operation of a radially polarized Nd:YAG laser using an annularly reflectivity-modulated photonic crystal coupler. Opt Lett. 2008 Oct;33(19):2278-80. Available from: <https://opg.optica.org/ol/abstract.cfm?URI=ol-33-19-2278>.
- [23] Anokhov S. A laser cavity used as an etalon for interferometric control of its own field. Optics & Laser Technology. 1998;30(1):23-6. Available from: <https://www.sciencedirect.com/science/article/pii/S0030399297000649>.
- [24] Ugolini D, Rafferty H, Winter M, Rockstuhl C, Bergmann A. LIGO analogy lab—A set of undergraduate lab experiments to demonstrate some principles of gravitational wave detection. American Journal of Physics. 2019 01;87(1):44-56. Available from: <https://doi.org/10.1119/1.5066567>.
- [25] Dimitrova TL, Weis A. Single photon quantum erasing: a demonstration experiment. European Journal of Physics. 2010 apr;31(3):625. Available from: <https://doi.org/10.1088/0143-0807/31/3/020>.
- [26] Barker DS, Reschovsky BJ, Pisenti NC, Campbell GK. Enhanced magnetic trap loading for atomic strontium. Physical Review A. 2015 Oct;92(4). Available from: <http://dx.doi.org/10.1103/PhysRevA.92.043418>.
- [27] CORNERSTONE. CORNERSTONE Standard Components Library (On 300 nm Si3N4 Platform). CORNERSTONE; 2022.
- [28] NIST/SEMATECH. What Are Control Charts?;. Available from: <https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc31.htm>.
- [29] Lohead G. Excited state spatial distributions in a cold strontium gas [Doctoral thesis]. Durham University; 2012. Available from: <https://etheses.durham.ac.uk/id/eprint/6329/>.
- [30] Goossen-Schmidt N, Arar B, Fricke J, Häusler K, Maaßdorf A, Schiemangk M, et al. DBR Lasers at 679.29 and 707.20 nm for Efficient Repumping in Sr Optical Lattice Clocks. IEEE Photonics Technology Letters. 2025;37(20):1205-8.
- [31] Vargas GR. A dual Mach-Zehnder interferometer wavelength measurement device using silicon over insulator technology. In: 2016 IEEE 7th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON); 2016. p. 1-4.

- [32] Yang B, Shi W, Pei C, Yu H, Jiang X, Hao Y, et al. Dual Mach-Zehnder Based Integrated X-Type Ratiometric Wavelength Monitor on Glass. *IEEE Photonics Technology Letters*. 2014 03;26:433-6.
- [33] Statistics How To. Lissajous Curves (Bowditch Curve);. Available from: <https://www.statisticshowto.com/lissajous-curves/>.
- [34] Poon JKS, Govdeli A, Sharma A, Mu X, Chen FD, Xue T, et al. Silicon photonics for the visible and near-infrared spectrum. *IEEE Journal of Selected Topics in Quantum Electronics*. 2023.
- [35] Thanh TTT, Hang DNT, Trung HN, Trinh MT, Tam LHD, Tan HN, et al. Design and Experimental Demonstration of a High-Performance 2×2 Restricted Interference MMI Coupler-Based Optical Power Splitter for C-Band Applications. *IEEE Photonics Journal*. 2022.
- [36] Nelan SP, Mercante A, Shi S, Yao P, Shahid E, Shopp BA, et al. Ultra-High Extinction Dual-Output Thin-Film Lithium Niobate Intensity Modulator. *IEEE Access*. 2021.
- [37] Soldano LB, Pennings ECM. Optical multi-mode interference devices based on self-imaging: principles and applications. *Journal of Lightwave Technology*. 1995;13(4):615-27.
- [38] Hayden J, Hugger S, Fuchs F, Lendl B. A quantum cascade laser-based Mach-Zehnder interferometer for chemical sensing employing molecular absorption and dispersion. *Applied Physics B*. 2018;124:29. Available from: <https://doi.org/10.1007/s00340-018-6899-8>.
- [39] Chrostowski L, Hochberg M. Silicon photonics design: from devices to systems. Cambridge University Press; 2015.
- [40] Muñoz P, Micó G, Bru LA, Pastor D, Pérez D, Doménech JD, et al. Silicon nitride photonic integration platforms for visible, near-infrared and mid-infrared applications. *Sensors*. 2019;19(11):2565.
- [41] CORNERSTONE. Silicon Nitride Visible MPW Design Guidelines; 2024. Available at: <https://www.cornerstone.sotonfab.co.uk>.